

Virtuoso[®] Layout Pro: T3 Basic Commands

Course Version IC 25.1

Lecture Manual

Revision 1.0

© 1990-2025 Cadence Design Systems, Inc. All rights reserved.

Printed in the United States of America.

Cadence Design Systems, Inc. (Cadence), 2655 Seely Ave., San Jose, CA 95134, USA.

Trademarks: Trademarks and service marks of Cadence Design Systems, Inc. (Cadence) contained in this document are attributed to Cadence with the appropriate symbol. For queries regarding Cadence trademarks, contact the corporate legal department at the address shown above or call 1-800-862-4522.

All other trademarks are the property of their respective holders.

Restricted Print Permission: This publication is protected by copyright, and any unauthorized use of this publication may violate copyright, trademark, and other laws. Except as specified in this permission statement, this publication may not be copied, reproduced, modified, published, uploaded, posted, transmitted, or distributed in any way without prior written permission from Cadence. This statement grants you permission to print one (1) hard copy of this publication subject to the following conditions:

- The publication may be used solely for personal, informational, and noncommercial purposes;

- The publication may not be modified in any way;

- Any copy of the publication or portion thereof must include all original copyright, trademark, and other proprietary notices, and this permission statement and

- Cadence reserves the right to revoke this authorization at any time, and any such use shall be discontinued immediately upon written notice from Cadence.

Disclaimer: Information in this publication is subject to change without notice and does not represent a commitment on the part of Cadence. The information contained herein is the proprietary and confidential information of Cadence or its licensors and is supplied subject to and may be used only by Cadence customers in accordance with a written agreement between Cadence and the customer.

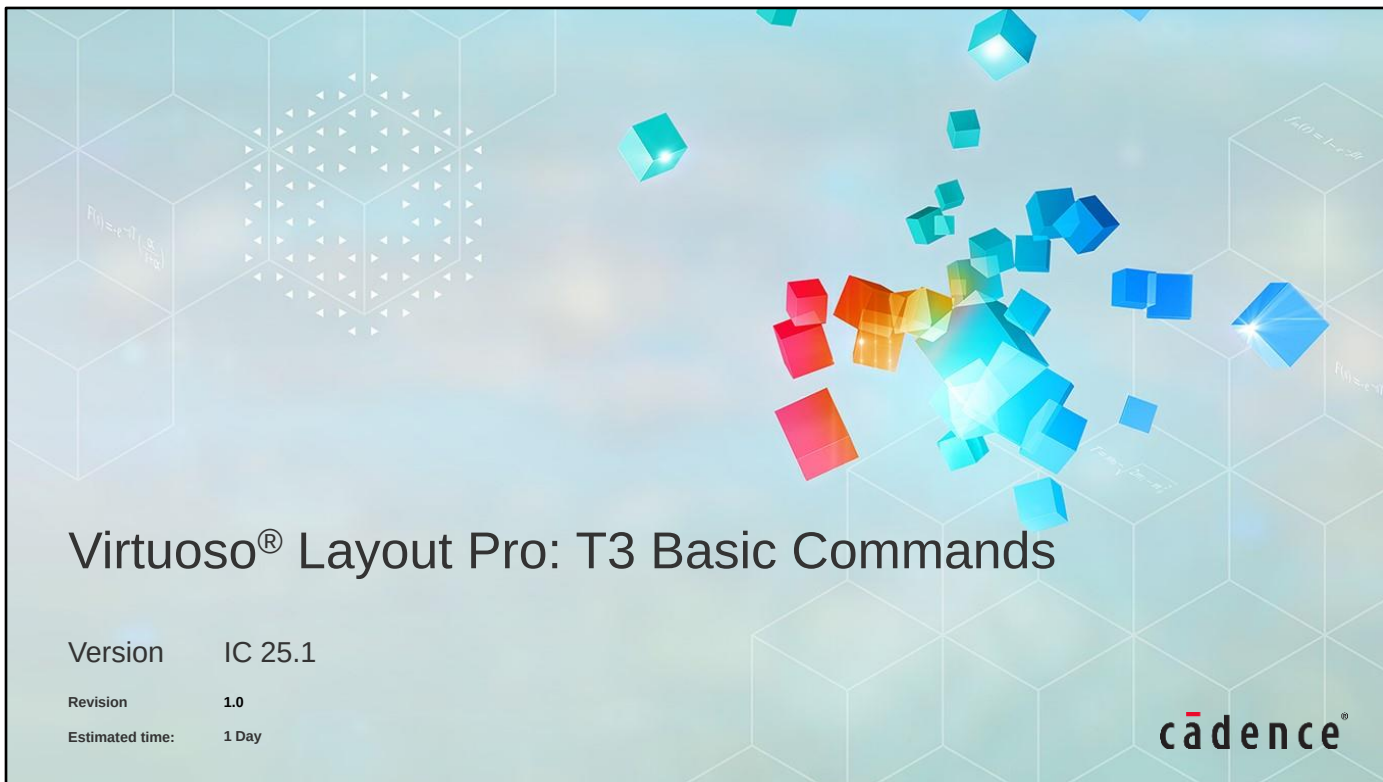
Except as may be explicitly set forth in such agreement, Cadence does not make and expressly disclaims any representations or warranties as to the completeness, accuracy, or usefulness of the information contained in this document. Cadence does not warrant that the use of such information will not infringe any third party rights, nor does Cadence assume any liability for damages or costs of any kind that may result from the use of such information.

Restricted Rights: Use, duplication, or disclosure by the Government is subject to restrictions as set forth in FAR52.227-14 and DFAR252.227-7013 et seq. or its successor.

Table of Contents

Virtuoso Layout Pro: T3 Basic Commands

Module 1	About This Course	2
	There are no labs in this module.	
Module 2	Design Creation and Placement	9
	Submodule Generate All From Source	12
	Submodule Generate Selected From Source	25
	Submodule Auto Place	29
	Submodule Virtuoso Custom Digital Placer	34
	Lab 2-1 Populating a Design with Generate All From Source	
	Lab 2-2 Populating a Design with Generate Selected From Source	
	Lab 2-3 Placing Components with the Auto Place	
	Lab 2-4 Placing Components with the Custom Digital Placer	
Module 3	Binder and Extractor.....	57
	Submodule XL Compliance	59
	Submodule Manual Binding	72
	Submodule Incremental Binding	77
	Submodule Update Binding	88
	Submodule Complex Binding	101
	Lab 3-1 Using Manual Binding	
	Lab 3-2 Using Incremental Binding	
	Lab 3-3 Using Update Binding	
	Lab 3-4 Using Complex Binding	
Module 4	Using Constraint Groups in Virtuoso.....	121
	Lab 4-1 Using Constraint Groups	
Module 5	Course Conclusions	135
Module 6	Next Steps	138



This page does not contain notes.



Module 1

About This Course

cādence®

This page does not contain notes.

Course Prerequisites

Before taking this course, you should have completed the following courses:

- Virtuoso® Layout Pro: T1 Environment and Basic Commands
- Virtuoso Layout Pro: T2 Create and Edit Commands

Or you need to have:

- Experience with layout design
- Knowledge of schematic symbols and MOS devices
- A basic knowledge of UNIX/Linux



This page does not contain notes.

Course Objectives

In this course, you

- Use Virtuoso Layout Suite EXL to create a new layout cell from scratch
- Debug and address connectivity errors in your layout design
- Analyze how the Layout EXL binder and extractor work
- Describe how the Constraint Groups affect working with VS-EXL



This page does not contain notes.

Course Agenda

- Design Creation and Placement
 - Populating a Design with Generate All From Source
 - Populating a Design with Generate Selected From Source
 - Placing Components with the Auto Place
 - Placing Components with the Custom Digital Placer
- Binder and Extractor
 - Using Manual Binding
 - Using Incremental Binding
 - Using Update Binding
 - Using Complex Binding
- Using Constraint Groups in Virtuoso
 - Using Constraint Groups



This page does not contain notes.

Software and Licenses

For the software and licenses used in the labs for this course, go to:

https://www.cadence.com/en_US/home/training/all-courses/85089.html

If there is additional information regarding the specific software, it is detailed in the lab document and/or the README file of the database provided with this course.



This page does not contain notes.

Become Cadence Certified by Earning a Digital Badge



Digital badges indicate mastery in a certain technology or skill and give managers and potential employers a way to validate your expertise.

- Cadence Training Services offers digital badges for our popular training courses.
- Your digital badge can be added to your email signature or social media platforms like LinkedIn or Facebook.

Benefits of Cadence Certified Digital Badges

- Validate expertise
 - Expand career opportunities
- Professional credibility
 - Stand apart from your peers
- For more information, go to www.cadence.com/training or email es_digitalbadge@cadence.com.



How do I register to take the exam?

- Log in to our [Learning Management System](#), click on the course in your transcript, and go to the **Content** tab to locate the exam.

How long will it take to complete the exam?

- Most exams take 45 to 90 minutes to complete. You may retake the exam multiple times to pass the exam.

How do I access and use the digital badge?

- After you pass the exam, you get a digital badge and instructions on how to place it on social media sites.

How is the digital badge validated?

- [Credly](#) validates the digital badge as issued to you by Cadence and includes the details of the criteria you completed to earn the badge.

This page does not contain notes.

Icons Used in This Class



Best Practice



Language Syntax



Concept/
Glossary



Frequently Asked
Questions /
Quiz



Error Message

Throughout this class, we use icons to draw your attention to certain kinds of information. Here are the icons we use, and what they mean.



Problem & Solution



Quick Reference



Tool Command



How To



Lab List

This page does not contain notes.



Module 2

Design Creation and Placement

cādence[®]

This page does not contain notes.

Module Objectives

In this module, you generate initial placements using:

- Generate All From Source (GFS)
- Generate Selected From Source (GSFS)
- Auto Place (AP)
- Virtuoso® Custom Digital Placer (VCP)



This page does not contain notes.

Terms and Definitions: Placer Commands



Generate All From Source (GFS)

Populates all pins and instances required in your layout automatically



Generate Selected From Source (GSFS)

Populates components yet unplaced from the schematic to the layout automatically



Auto Place (AP)

Automatically places the layout objects using the Auto Place



Virtuoso Custom Digital Placer (VCP)

Automatically places standard cells of the block within rows

- GFS – Generate All From Source
- GSFS – Generate Selected From Source
- AP –Auto Place
- VCP – Virtuoso Custom Digital Placer



Submodule 2-1

Generate All From Source

cādence®

This page does not contain notes.

Submodule Objective

In this submodule, you generate initial placements using:

- Generate All From Source (GFS)



This page does not contain notes.

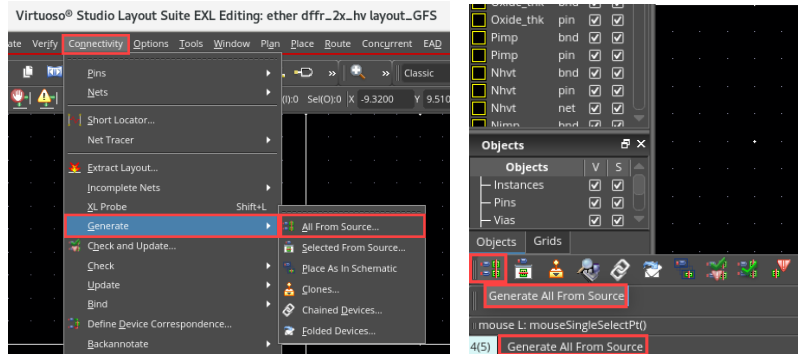
Generating a Layout from GFS



Generate all from the source by choosing either the **Connectivity – Generate – All From Source** menu command or by clicking the **Generate All From Source (GFS)** icon.



- Use the GFS command to populate the design elements from the schematic to an empty or partially populated layout canvas.



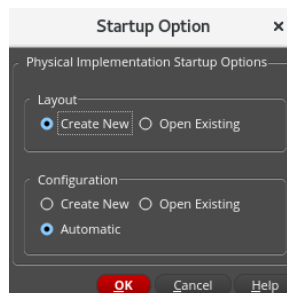
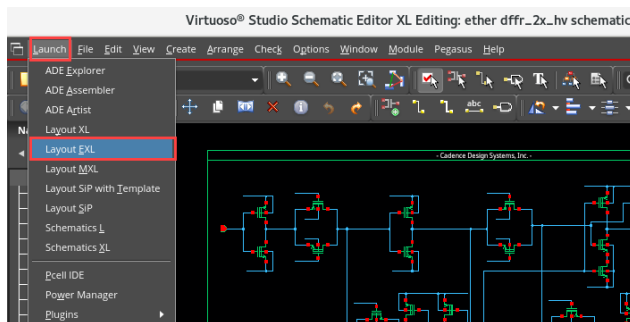
Use the *Generate All From Source* command to generate layout representations of the schematic design components. *Generate All From Source* deletes any existing components in the layout view and generates everything from scratch.

Notice that you can use an ever-increasing number of commands through the use of icons instead of the pull-down menus. You can hover over each icon to see the command mapped to it before clicking on it.

Generating a Layout from Source



To create a new layout from the source, choose **Launch – Layout EXL** in the schematic, and select **Create New** option in the Startup Option form.



- Use the Startup Option form to control the Layout EXL launch operation.
- Depending on what you specify, you will be asked to complete different forms as Layout EXL is launched.

15 © Cadence Design Systems, Inc. All rights reserved.

cadence

Use the Startup Option form to control the Layout EXL launch operation. Depending on what you specify, you will be asked to complete different forms as Layout EXL is launched.

- **Layout** controls whether the system creates a new layout view or opens an existing one for the schematic.
- **Create New** opens the New File form where you type the name of a new layout view for your Layout EXL design. For more information, see *Managing Libraries in the Cadence® Library Manager User Guide*.
- **Open Existing** opens the Open File form where you can choose an existing layout view. For more information on this form, see *Working with Cellviews in the Virtuoso Design Environment User Guide*.
- **Configuration** controls the physical configuration view used for your design.
- **Create New** opens the Create Physical Configuration View form where you can specify a new physical configuration view for the design. (This happens after you have specified the layout view.)
- **Open Existing** opens the Open Physical Configuration View form where you can choose an existing physical configuration view. (This happens after you have specified the layout view.)
- **Automatic** tells Layout XL to handle all operations related to the physical configuration view automatically.

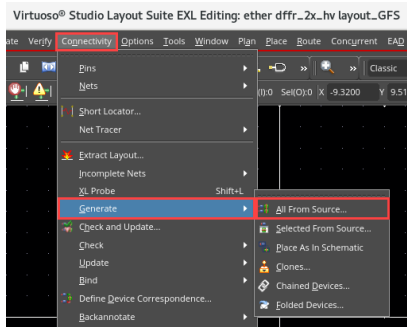
The system creates a physical configuration view (or opens it, if one already exists) containing the minimum information required for Layout XL to operate correctly and manages this information automatically throughout the session.

If you need to modify the hierarchy configuration, component type definitions, or soft blocks at a later time, use the *Launch – Configure Physical Hierarchy* command. For more information, see *Configuring the Physical Hierarchy in Virtuoso Layout Suite XL, EXL User Guide, Product Version IC23.1*.

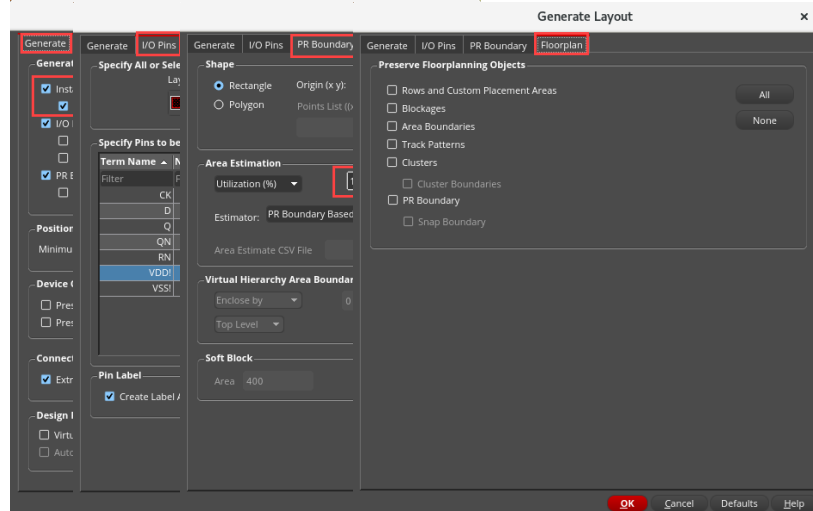
Generate All From Source: Interface



When the blank layout canvas opens, choose the **Connectivity – Generate – All From Source** command. The tabbed Generate Layout form appears.



- Use the Generate Layout form to generate layout representations of schematic design components.



You can display the Generate Layout form by choosing the **Connectivity – Generate – All from Source** pull-down menu entry or by clicking the **Generate All From Source** Layout XL Toolbar icon at the lower-left corner of the layout canvas.

Generate Layout

Use the Generate Layout form to generate layout representations of schematic design components.

The form is split into four tabs:

- Generate
- I/O Pins
- PR Boundary
- Floorplan

Generate All From Source: Generate Tab

The screenshot shows the 'Generate Layout' dialog box with the 'Generate' tab selected. The dialog is divided into several sections: 'Generate', 'Position', 'Device Correspondence', 'Connectivity Extraction', and 'Design Planning'. Callouts point to specific options: 'Instances' (checked), 'Chain' (checked), 'I/O Pins' (checked), 'PR Boundary' (checked), 'Snap Boundary' (checked), 'Minimum Separation' (set to 1), 'In Boundary' (checked), 'Preserve User-Defined Bindings' (unchecked), 'Preserve physConfig Data in Layout' (unchecked), 'Extract Connectivity after Generation' (checked), 'Virtual Hierarchy' (unchecked), 'Auto Generate Soft Blocks' (unchecked), and 'As Pcells' (unchecked).

Generates all the instances in the schematic that do not have one of the ignore properties attached to them.

Generates all the pins listed on the I/O Pins tab.

Generates a place-and-route boundary based on the settings on the PR Boundary tab.

Runs connectivity extraction as part of the layout generation process. Check this option to see the incomplete nets in the design immediately after layout generation has run.

Positions the instances in the layout at a minimum separation based on the value you specify.

Generates layout representations within the design boundary when the Generate All From Source command is run.

Preserves user-defined bindings of devices between the schematic and the layout.

17 © Cadence Design Systems, Inc. All rights reserved.

cadence

You use the **Generate** tab to determine whether or not to generate instances, do automatic chaining and folding, and include I/O pins and boundaries when you run **Generate All From Source**.

You also determine whether or not to automatically extract the physical design when you launch Gen From Source.

- **Minimum Separation** – Positions the instances in the layout at a minimum separation based on the value you specify. You can also control this option using the *lxPositionMinSep* environment variable.
- **In Boundary** – Generates layout representations within the design boundary when the Generate All From Source command is run. You can also control this option using the *lxGenerateInBoundary* environment variable.
- **Preserve User-Defined Bindings** – Preserves user-defined bindings of devices between the schematic and the layout. This option preserves only user-defined one-to-one, many-to-many, many-to-one, and one-to-many device correspondence defined in the Define Device Correspondence form. It does not report missing devices or shapes within a bound group.
- **Extract Connectivity after Generation** – Runs connectivity extraction as part of the layout generation process. Check this option to see the incomplete nets in the design immediately after layout generation has run.

See complete details on the use of this form by clicking **Help** in the lower-right corner of the form.

Generate All From Source: I/O Pins Tab

Specifies attribute values and applies them to all the pins shown in the list box.

Selects pins from the list box and updates the attribute values used when those pins are generated in the layout.

Specifies the type of label generated when you create a pin. Honored by the Generate All From Source and Generate Selected From Source commands.

Term Name	Net Name	Layer	Width	Height	Num
CK	CK	Metal1 drawing	0.12	0.12	1
D	D	Metal1 drawing	0.12	0.12	1
Q	Q	Metal1 drawing	0.12	0.12	1
QNI	QNI	Metal1 drawing	0.12	0.12	1
RNI	RNI	Metal1 drawing	0.12	0.12	1
VDDI	VDDI	Metal2 drawing	1	1	1
VSSI	VSSI	Metal2 drawing	1	1	1



You use the I/O Pins tab to determine which layers your pins acquire when you run **Generate All From Source**.

You also determine pin size, number per net, and whether or not to generate labels for pins.

See complete details on the use of this form by clicking **Help** in the lower-right corner of the form.

Generate All From Source: PR Boundary Tab

Specifies whether the place and route boundary is a rectangle or a polygon.

Specifies how the system calculates the shape and size of a rectangular boundary.

Generate Layout

Generate | I/O Pins | **PR Boundary** | Floorplan

Shape

☒ Rectangle Origin (x y): (0.0 0.0)
☐ Polygon Points List (x1 y1 x2 y2...)

Area Estimation

Utilization (%) 15 Aspect Ratio (W/H) 1
Estimator: PR Boundary Based Register...
Area Estimate CSV File

Virtual Hierarchy Area Boundary

Enclose by 0.5
Top Level

Soft Block

Area 400

OK Cancel Defaults Help

19 © Cadence Design Systems, Inc. All rights reserved.



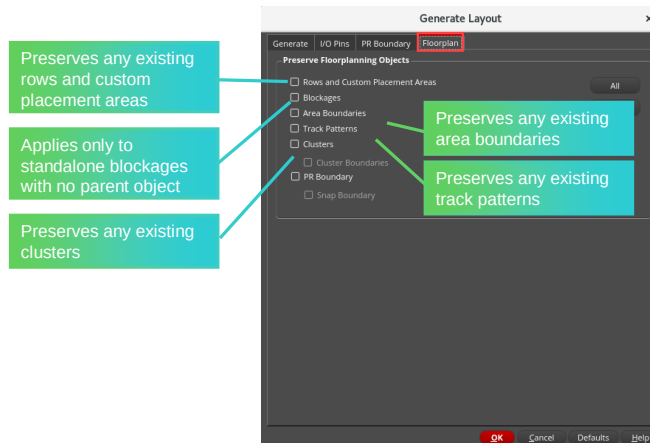
You use the PR Boundary tab to determine the shape and size of the boundary when you run **Generate All From Source**.

You can also let the software determine the approximate size of the boundary by choosing **Utilization (%)** under the Area Estimation pull-down in conjunction with the **Aspect Ratio (W/H)** option if you are not familiar with the boundary size.

If you already know the boundary size, you can use the Width and Length options.

See complete details on the use of this form by clicking **Help** in the lower-right corner of the form.

Generate All From Source: Floorplan Tab



20 © Cadence Design Systems, Inc. All rights reserved.



You use the Floorplan tab to preserve floorplanning objects when you run **Gen From Source**.

- *Rows and Custom Placement Areas* – Preserves any existing rows and custom placement areas.
- *Blockages* – Apply only to standalone blockages which have no parent object. A blockage with a parent object is preserved automatically along with its parent.
- *Area Boundaries* – Preserves any existing area boundaries.
- *Track Patterns* – Preserves any existing track patterns.
- *Clusters* – Preserves any existing clusters.
 - *Cluster Boundaries* – Only if clusters are also preserved.

See complete details on the use of this form by clicking **Help** in the lower-right corner of the form.

Virtuoso Layout Suite XL: lxGenerateInBoundary

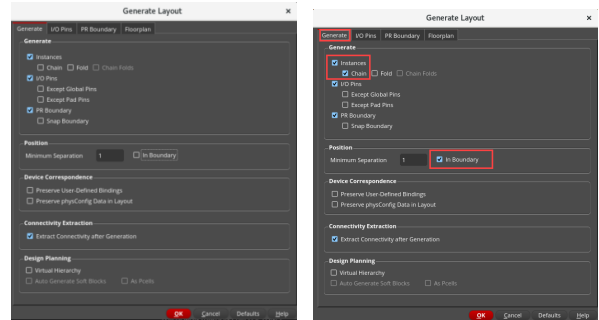


Add/Enter in `.cdsinit` file/CIW: `envSetVal("layoutXL" "lxGenerateInBoundary" 'boolean { t | nil } )`

OR

Add in `.cdsenv` file: `layoutXL lxGenerateInBoundary boolean { t | nil }`

- `envSetVal("layoutXL" "lxGenerateInBoundary" 'boolean t) :`
Generated components are placed within the design boundary when GFS is run.
- `envSetVal("layoutXL" "lxGenerateInBoundary" 'boolean nil) :`
Generated components are placed below the design boundary when GFS is run.



21 © Cadence Design Systems, Inc. All rights reserved.

cadence

You have the option to place all the generated components in or out of the *prBoundary* in the Generate All From Source (Generate Layout) form in the Generate tab under the Position section.

Environment Variable

Add/Enter in `.cdsinit` file/CIW: `envSetVal("layoutXL" "lxGenerateInBoundary" 'boolean { t | nil } )`

OR

Add in `.cdsenv` file: `layoutXL lxGenerateInBoundary boolean { t | nil }`

- `t` Within the design boundary
- `nil` Below the design boundary

Add this environment variable in your `.cdsinit/.cdsenv` file.

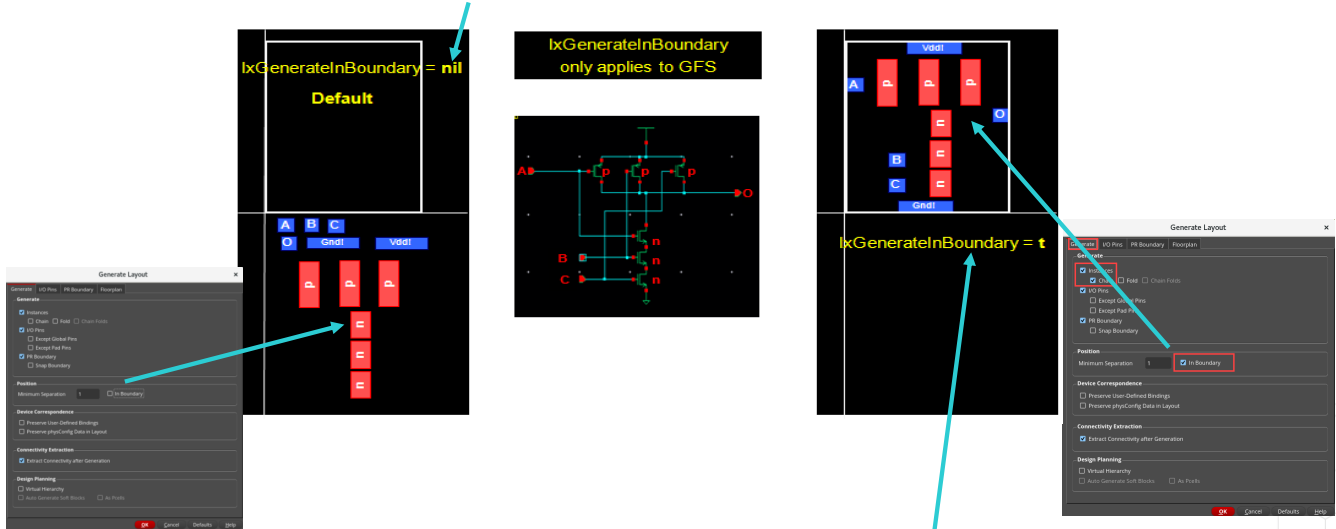
Generates layout representations within the design boundary when the Generate All From Source command is run. The default is `nil`.

Examples

```
envGetVal("layoutXL" "lxGenerateInBoundary")
envSetVal("layoutXL" "lxGenerateInBoundary" 'boolean t)
envSetVal("layoutXL" "lxGenerateInBoundary" 'boolean nil)
```


Placement Results: *lxGenerateInBoundary*

Shows the *GFS* placement result when *lxGenerateInBoundary* is *nil*.



Shows the *GFS* placement result when *lxGenerateInBoundary* is *t*.

Placement Results of *lxGenerateInBoundary* are shown here.

GFS placement result when *lxGenerateInBoundary* is *t/nil* is shown.

Virtuoso Layout Suite XL: lxPositionMinSep

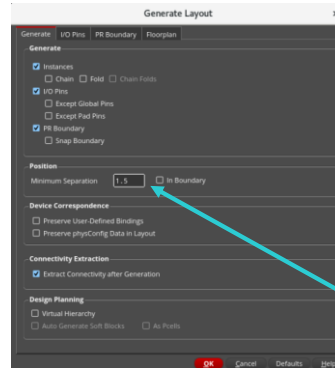
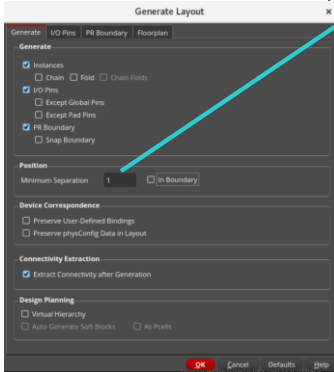


Add/Enter in `.cdsinit` file/CIW: `envSetVal("layoutXL" "lxPositionMinSep" 'float float)`

OR

Add in `.cdsenv` file: `layoutXL lxPositionMinSep float float`

The default value of `lxPositionMinSep` is 1.



`envSetVal("layoutXL" "lxPositionMinSep" 'float 1.5)` → Positions the instances in the layout with 1.5 separation value

23 © Cadence Design Systems, Inc. All rights reserved.

cadence

- Positions the instances in the layout at a minimum separation based on the user-defined value

Environment Variable

Add/Enter in `.cdsinit` file/CIW: `envSetVal("layoutXL" "lxPositionMinSep" 'float float)`

OR

Add in `.cdsenv` file: `layoutXL lxPositionMinSep float float`

The environment variable can be used to spread out or compact the instances in a layout by increasing or decreasing the minimum separation value.

Note: The `lxPositionMinSep` environment variable takes precedence over `lxGenerateInBoundary`. Therefore, if the minimum separation value you specify using `lxPositionMinSep` is large, the instances may get positioned outside of the design boundary even if `lxGenerateInBoundary` is set.

- Used by the following commands: *Generate All From Source*, *Update Components And Nets*, *Pick From Schematic*, *Place As In Schematic*, *Generate Folded Devices*

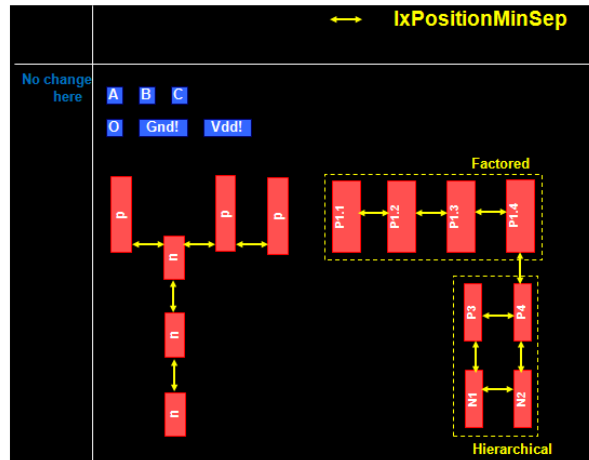
Examples

```
envGetVal("layoutXL" "lxPositionMinSep")
```

```
envSetVal("layoutXL" "lxPositionMinSep" 'float 1.5)
```

Placement Results: *lxPositionMinSep*

- The default value of *lxPositionMinSep* is 1.
- Specifying a greater separation can throw instances outside the *prBoundary*.



24

© Cadence Design Systems, Inc. All rights reserved.



Placement Results of *lxPositionMinSep* are shown here.

The default value of *lxPositionMinSep* is 1.

Specifying a greater separation can throw instances outside the *prBoundary*.



Submodule 2-2

Generate Selected From Source

cā dence®

This page does not contain notes.

Submodule Objective

In this submodule, you generate initial placements using:

- Generate Selected From Source (GSFS)



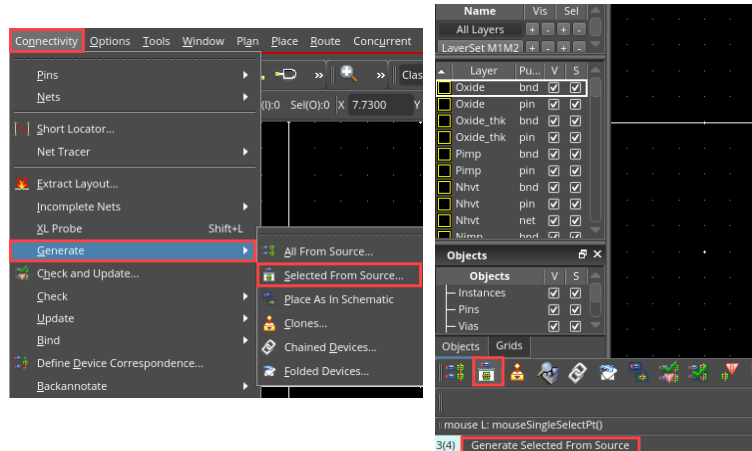
This page does not contain notes.

Generating a Layout from GSFS



Generate selected from the source by choosing either **Connectivity – Generate – Selected From Source** menu command or by clicking the **Generate Selected From Source (GSFS)** icon.

- Use the GSFS command to populate the design elements from the schematic to an empty or partially populated layout canvas.



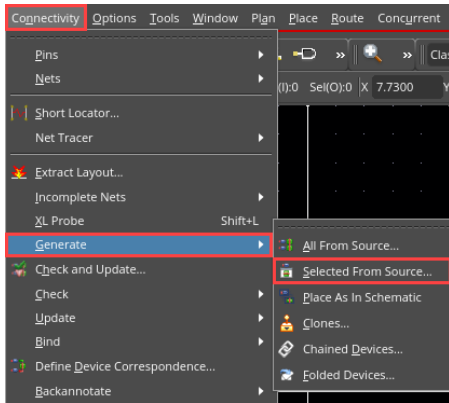
Use the Generate Selected From Source command to manually generate schematic instances and pins in the layout view. You can select the components to be generated either before or after you launch the command and then generate the components either all together or one at a time in the layout view.

Notice that you can use an ever-increasing number of commands through the use of icons, instead of the pull-down menus. You can hover over each icon to see the command mapped to it before clicking on it.

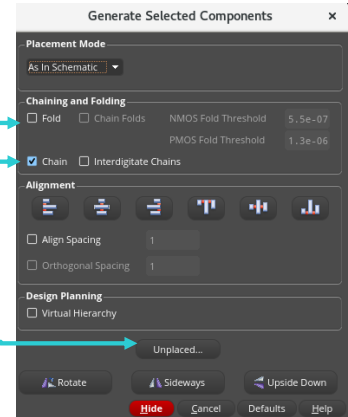
Generate Selected From Source: Interface



When the blank layout canvas opens, choose the **Connectivity – Generate – Selected From Source** command. The Generate Selected Components form appears.



Populating components this way will allow you to pick individual or multiple components from the schematic before placing them in your layout, as well as ordering chaining or folding. You can also see and select any Unplaced components by clicking the **Unplaced** button.



You can display the Generate Selected Components form by choosing the **Connectivity – Generate – Selected From Source** pull-down menu entry or by clicking the **Generate Selected From Source** Layout EXL Toolbar icon at the lower-left corner of the layout canvas.

Generate Selected Components

Use the Generate Selected Components form to generate selected schematic instances and pins directly in the layout view.



Submodule 2-3

Auto Place

cādence®

This page does not contain notes.

Submodule Objective

In this submodule, you generate initial placements using:

- Auto Place (AP)



This page does not contain notes.

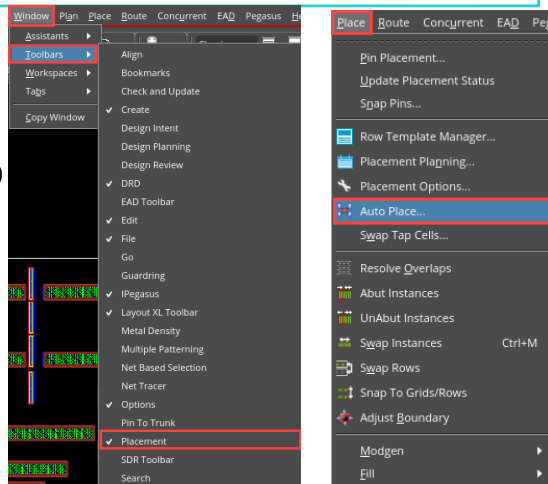
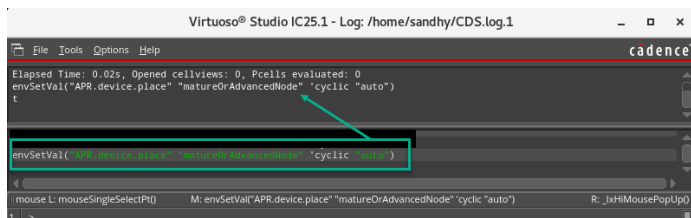
Invoking the Auto Place on Your Palette



You can invoke the Auto place by choosing **Window – Toolbar – Placement**, or choose the **Place – Auto Place**.



- Use the Auto Place / Analog auto placer command to place the components from your schematic to the layout canvas quickly.
- In the CIW window, enter the below command.
`envSetVal("APR.device.place" "matureOrAdvancedNode" 'cyclic "auto")`



You can invoke the Auto Place / Analog Auto Placer by choosing the Window – Toolbars – Placemenet or Go to Place - Auto_Place

You can use the Analog Auto Placer command to place the components from your schematic to the layout canvas quickly.

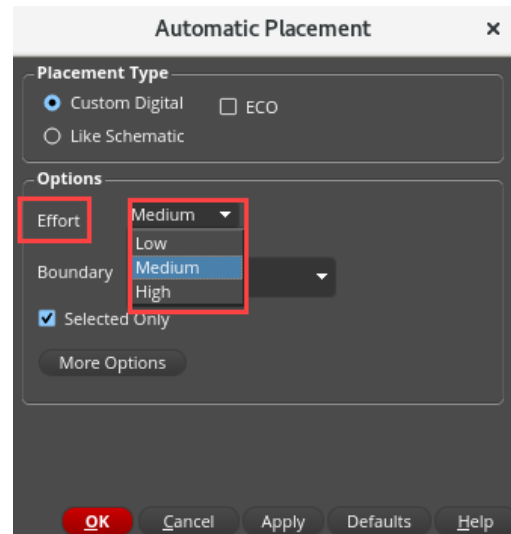
n the CIW window enter the below command.

`envSetVal("APR.device.place" "matureOrAdvancedNode" 'cyclic "auto")`

Refer to the *Virtuoso Auto P &R User Guide, Product Version IC25.1*, for more details.

Invoking the Auto Place on Your Palette (continued)

- There are three Placement Effort Level options for Auto Place:
 - Low
 - Medium
 - High



Creating an Automatic Placement

You can create an automatic placement. Like quick placement, full automatic placement respects constraints set in the Constraint Manager. In addition, it also optimizes for area or wire length and avoids design rule violations.

Environment Variable: aapDefaultEffort

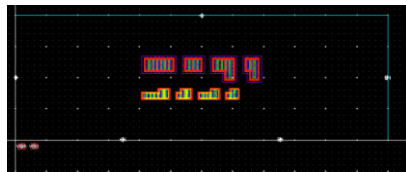
See complete details on the use of this form by clicking **Help** in the lower-right corner of the form.

Generate From Source and Auto Place: Examples



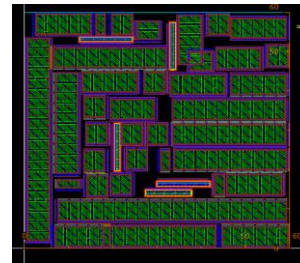
GFS Example

All the components are auto placed within the **prBoundary** with the **In Boundary** option ON.



Auto Place Example

Use the Automatic Placement form to select the type of placement required. Set the corresponding options, Field Description Placement Type. This group box specifies the placement options for digital and analog. Related Topics: Running the Automatic Place.



Generate From Source and *Auto Place* examples are shown here.

In the *Generate From Source* example shown here, all the components are auto placed within the *prBoundary* with the *In Boundary* option on.

In the *Automatic Placement* example shown here,

Use the *Automatic Placement* form to select the type of placement required set the corresponding options Field Description Placement Type This group box specifies the placement options for digital and analog Related Topics Running the Automatic Placer Setting the Assisted Placement Options in the Placement Options Form Setting Automatic Placement Options in the Placement Options Form Setting Common Options in the Placement Options Form



Submodule 2-4

Virtuoso Custom Digital Placer

cā dence®

This page does not contain notes.

Submodule Objective

In this submodule, you generate initial placements using:

- Virtuoso® Custom Digital Placer (VCP)



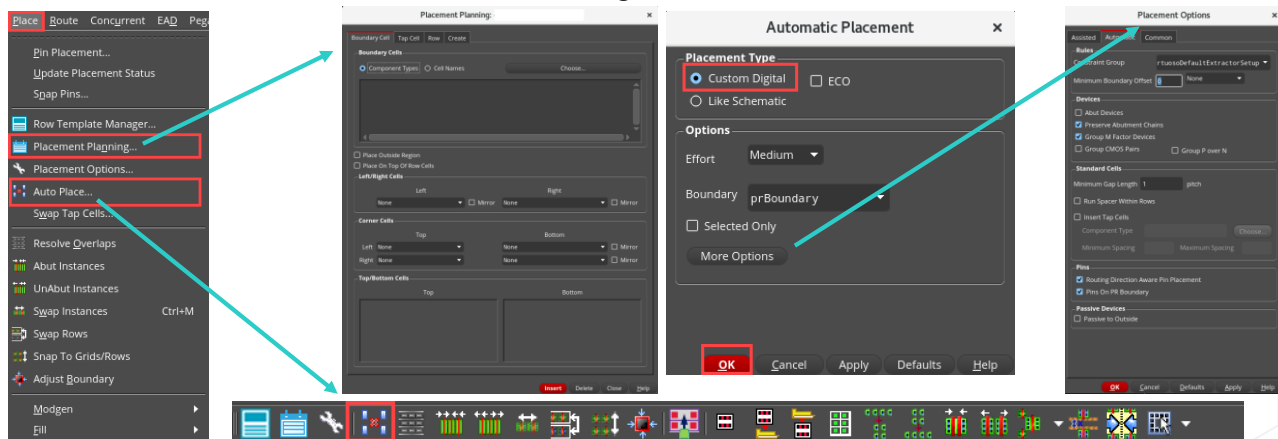
This page does not contain notes.

Activating the Placement Toolbar: VCP



You can activate the *Placement* toolbar from the **Window – Toolbars – Placement** menu in your layout window. **Place – Auto Place**

The Placement toolbar has **Placement Planning** icons and **Automatic Placement** icons.



36 © Cadence Design Systems, Inc. All rights reserved.

cadence

You can activate the Placement toolbar from the **Window – Toolbars – Placement** menu in your layout window if it is not already displayed.

The Placement toolbar has the Placement Planning and *Automatic Placement*.

Placement Planning and Automatic placement menus are also shown here.

See complete details on the use of these forms by clicking the Help button in the lower-right corner of these forms.

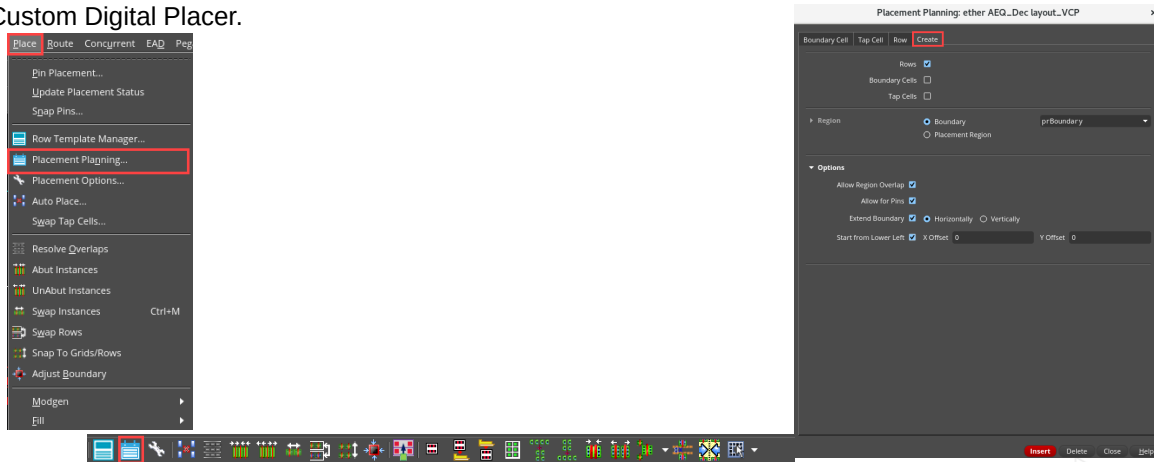
Invoking the Placement Planning Form on Your Palette



You can invoke the Placement Planning form by choosing **Place – Placement Planning** menu command or by clicking the **Placement Planning** icon.



- Use the **Placement Planning** form to prepare your design for placement before you run the Virtuoso® Custom Digital Placer.



37 © Cadence Design Systems, Inc. All rights reserved.

cadence

To plan a placement, choose Place –Placement Planning or click the **Placement Planning** icon. The Placement Planning form is displayed.

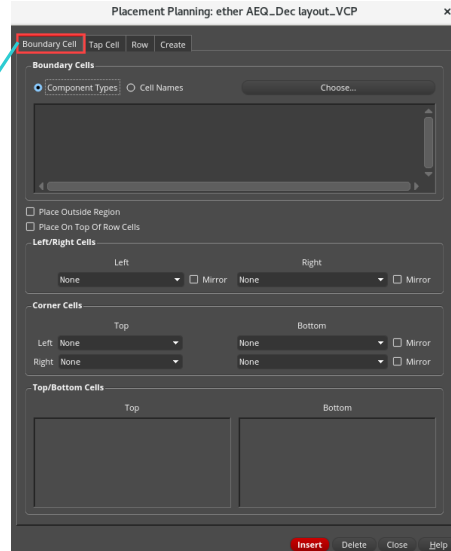
Use the Placement Planning form to prepare your design for placement before you run the Virtuoso® Custom Digital Placer.

Refer to the *Virtuoso Placement Planning User Guide, Product Version IC25.1* for more details.

Placement Planning: Create Boundary Tab

Specify whether you want to select boundary cells based on their Component Types or Cell Names, before running Custom Digital Placer.

Use the Boundary Cell tab (optional) to specify the information about the boundary cells to be placed around a set of rows.



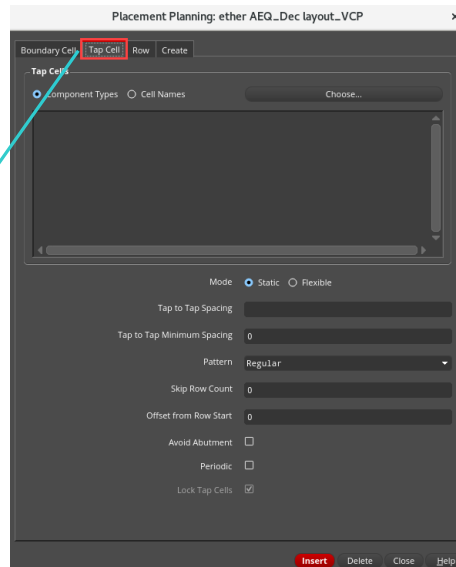
Specify whether you want to select boundary cells based on their Component Types or Cell Names. Click Choose to display the Select Cells form. The options in the form differ depending on whether Component Types or Cell Names was selected.

Refer to the *Virtuoso Placement Planning User Guide, Product Version IC25.1* for more details.

Placement Planning: Tap Cell Tab

Use the options on the **Tap Cell** tab of the Placement Planning form to insert tap cells before running Custom Digital Placer.

You can (optionally) insert either single-height or multi-height tap cells in the empty spaces between the standard cells.



39 © Cadence Design Systems, Inc. All rights reserved.



Use the options on the *Tap Cell* tab of the Placement Planning form to insert tap cells before running Custom Digital Placer.

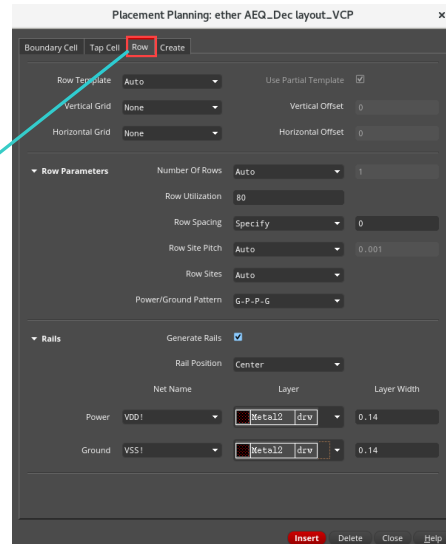
You can (optionally) insert either single-height or multi-height tap cells in the empty spaces between the standard cells. Tap cells are a set of contacts that are used to reduce latch-up effects between power and ground connections and the connections with wells or substrate contacts.

Tap cells are represented by cells that have their component class set to STDSUBCONT. Component types are defined in the Cells table of the Configure Physical Hierarchy form - Component Types mode. After ensuring that the tap cell definitions are in place, you can start inserting them between the core cells in your design.

Placement Planning: Row Tab

Use the options on the **Row** tab to define the row settings before generating them in the placement region.

Row-based placement provides tremendous advantage over the conventional placement methods.

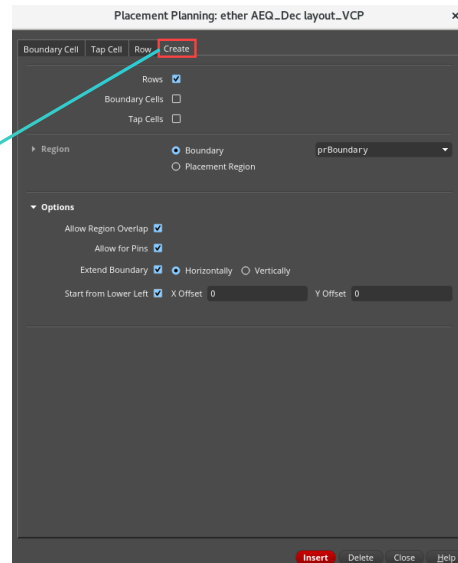


Row-based placement provides tremendous advantage over the conventional placement methods. Row-based placement helps improve routability, achieve better wire length after routing, and resolve illegal placements by balancing the number of instances placed in each row. It also helps resolve overlaps between the devices in each row. Use the options on the *Row* tab to define the row settings before generating them in the placement region.

Placement Planning: Create Tab

Use the options on the **Create** tab to define the generic settings.

Use the Create tab to set other placement parameters.



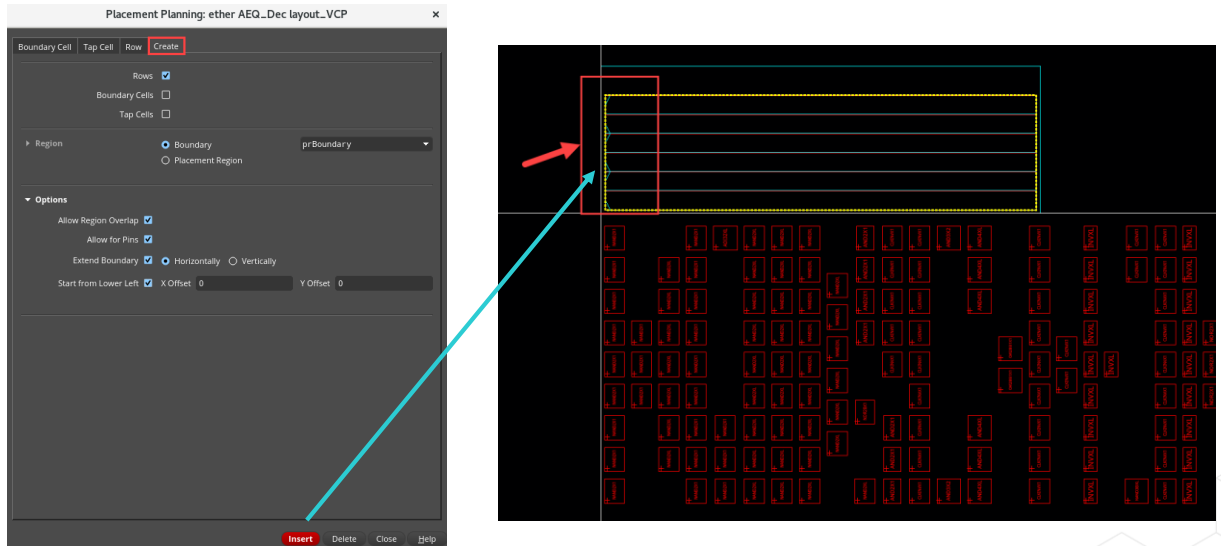
41 © Cadence Design Systems, Inc. All rights reserved.



After defining the *Tap Cell* and *Row* parameters, the final step is to define the generic placement settings before running the command. Use the options on the *Create* tab to define the generic settings.

Rows Generated

Click **Insert** to generate the specified tap cells and rows in the layout canvas.



42 © Cadence Design Systems, Inc. All rights reserved.



Click **Insert** to generate the specified tap cells and rows in the layout canvas.

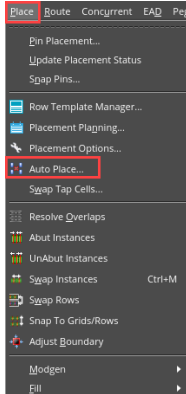
Invoking the Custom Digital Placer (VCP) Form on Your Palette



You can invoke the *Virtuoso Custom Digital Placer (VCP)* form by choosing **Place – Auto Place** menu command or by clicking the **Auto Place** icon.



- Use the Custom Digital Placer to place the transistors and other design elements automatically in your layout view.



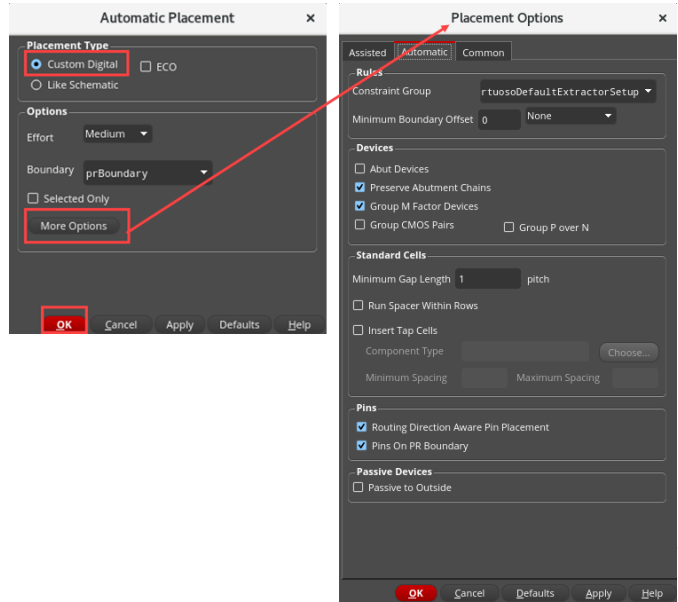
You can invoke the Virtuoso Custom Digital Placer (VCP) form by choosing the **Place –Auto Place** menu command or by clicking the **Auto Place** icon.

You can use the Custom Digital Placer to place the transistors and other design elements automatically in your layout view.

Refer to the *Virtuoso Auto Place User Guide, Product Version IC23.1*, for more details.

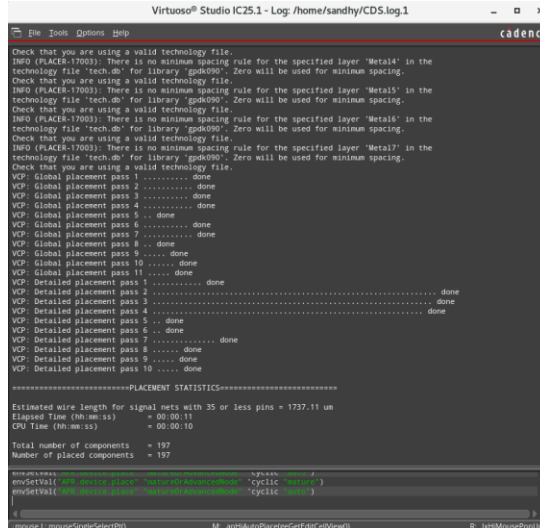
Using the Auto Placer Form

- Use the Automatic Placement form to define the type of placement that is run, the technology rules that are used, any automatic spacing adjustments to be done, and how the modified cellview is saved.



You can use the Automatic Placement form to define the type of placement that is run, the technology rules that are used, any automatic spacing adjustments to be done, and how the modified cellview is saved.

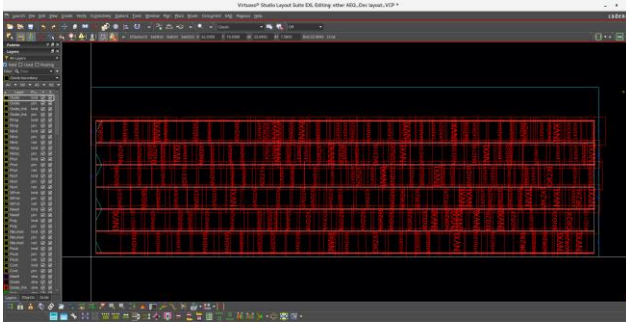
- Run VCP to automatically place standard cells of the block within rows.
 - VCP runs and the VLS-EXL Placement Status form opens as shown here:



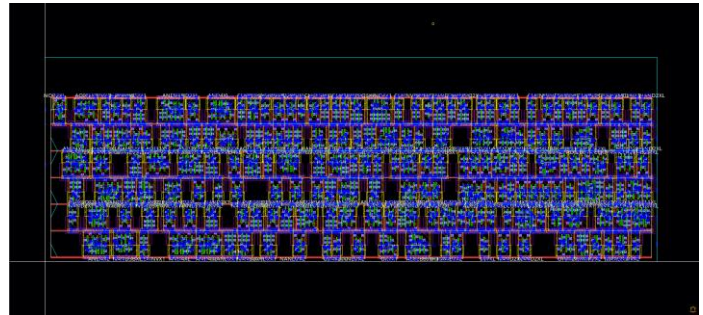
VCP runs and the VLS-EXL Placement Status form opens as shown.

Placement Result: Auto Placer

- The Virtuoso Custom Digital Placer has placed the components within rows.
 - The Auto Placer has also placed the pins because the *Allow For Pins* option is selected in the Placement Planning form.



Press the bindkey **Ctrl+F** to view the top level view in the layout.



Press the bindkey **Shift+F** to view all the levels of the hierarchy in the layout.



The Virtuoso Custom Digital Placer has placed the components within rows.

The Auto Placer has also placed the pins because the *Allow For Pins* option is selected in the Placement Planning form.

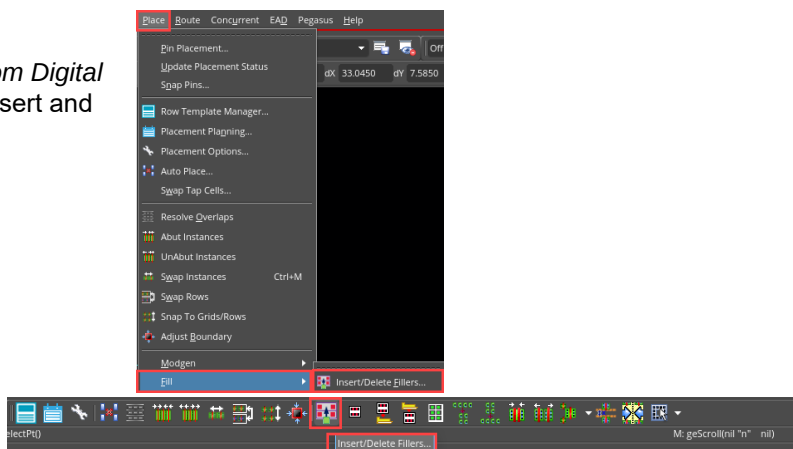
Invoking the Insert/Delete Filler Cells Form on Your Palette



You can invoke the Insert/Delete Filler Cells form by choosing **Place – Fill – Insert/Delete Fillers** menu command or by clicking the **Insert/Delete Fillers** icon.



- Use the *Virtuoso Custom Digital Placer's* capability to insert and delete filler cells.



Let us see how to invoke the Insert/Delete Filler Cells form on your Palette.

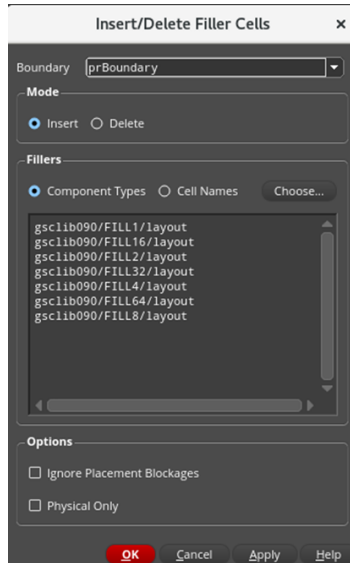
You can invoke the Insert/Delete Filler Cells form by choosing the **Place – Custom Digital – Insert/Delete Fillers** menu command or by clicking the **Insert/Delete Fillers** icon.

You can use the *Virtuoso Custom Digital Placer's* capability to insert and delete filler cells.

Refer to the *Virtuoso Custom Digital Placer User Guide, Product Version IC25.1* for more details.

Using the Insert/Delete Filler Cells Form

- Use the **Insert/Delete Filler Cells** form to either insert filler cells into a design or delete filler cells from a design.



The screenshot shows the 'Insert/Delete Filler Cells' dialog box. It has a title bar with a close button (X). The 'Boundary' dropdown is set to 'prBoundary'. The 'Mode' section has two radio buttons: 'Insert' (selected) and 'Delete'. The 'Fillers' section has two radio buttons: 'Component Types' (selected) and 'Cell Names', followed by a 'Choose...' button. Below these is a list box containing the following text: 'gsc1ib090/FILL1/layout', 'gsc1ib090/FILL16/layout', 'gsc1ib090/FILL2/layout', 'gsc1ib090/FILL32/layout', 'gsc1ib090/FILL4/layout', 'gsc1ib090/FILL64/layout', and 'gsc1ib090/FILL8/layout'. The 'Options' section has two checkboxes: 'Ignore Placement Blockages' and 'Physical Only', both of which are unchecked. At the bottom are four buttons: 'OK' (highlighted in red), 'Cancel', 'Apply', and 'Help'.

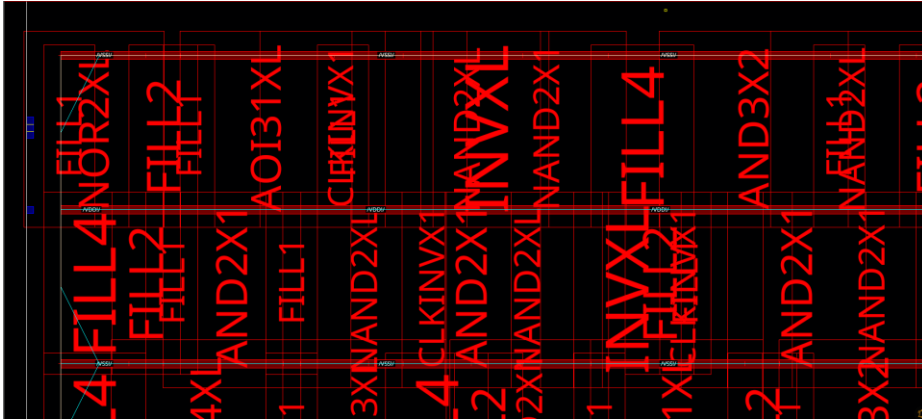
48 © Cadence Design Systems, Inc. All rights reserved.



You can use the Insert/Delete Filler Cells form to either insert filler cells into a design or delete filler cells from a design.

Insert/Delete Filler Cells Command Placement Result

- The *Insert/Delete Filler Cells* command has filled all the empty spaces between the placed cells with filler cells.
 - Zoomed in view of the layout displaying filler cells occupying the empty spaces between the placed components is shown here:



The Insert/Delete Filler Cells command has filled all the empty spaces between the placed cells with filler cells.

Zoomed in view of the layout displaying filler cells occupying the empty spaces between the placed components is shown on the slide.

Module Summary

In this module, you learned how to generate initial placements using:

- Generate All From Source (GFS)
- Generate Selected From Source (GSFS)
- Auto Place (AP)
- Virtuoso Custom Digital Placer (VCP)



This page does not contain notes.

Labs



Lab 2-1 Populating a Design with Generate All From Source

Lab 2-2 Populating a Design with Generate Selected From Source

Lab 2-3 Placing Components with the Auto Place

Lab 2-4 Placing Components with the Custom Digital Placer



This lab uses the Cadence 90nm Ethernet-Phy design.

Quiz: Populating the Design Elements from Schematic to Layout



What are the menu commands that you can use to populate the design elements from the schematic to an empty or partially populated layout canvas?

(Select all the correct choices.)

- A. Generate All From Source
- B. Generate Selected From Source
- C. Virtuoso Custom Digital Placer
- D. Analog Auto Placer
- E. All of the above

Answers: A and B

You can use *Generate All From Source* and *Generate Selected From Source* menu commands to populate the design elements from the schematic to an empty or partially populated layout canvas.



This page does not contain notes.

Quiz: Environment Variable to Place All the Generated Components In or Out of the prBoundary in the Layout Canvas



What is the environment variable that you can use to place all the generated components in or out of the *prBoundary* in the layout canvas? _____

(Enter the answer in the blank field.)

Answer: *lxGenerateInBoundary*

You can use *lxGenerateInBoundary* environment variable to place all the generated components in or out of the *prBoundary* in the layout canvas.



This page does not contain notes.

Quiz: Environment Variable to Position the Instances in the Layout Based on the User-Defined Value



The environment variable *lxPositionMinSep* positions the instances in the layout at a minimum separation based on the user-defined value.

(Select the single correct choice from the two choices.)

- A. True
- B. False

Answer: A

True. The environment variable *lxPositionMinSep* positions the instances in the layout at a minimum separation based on the user-defined value.



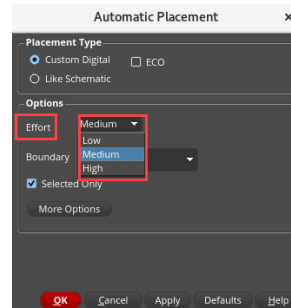
This page does not contain notes.

Quiz: Placement Effort Level in the Analog AutoPlace Form



What are the Placement Effort Level options available in the Automatic Placement form?
(Select all the correct choices.)

- A. Low
- B. Moderate
- C. Medium
- D. High
- E. Standard
- F. All of the above



Answers: A, C, and D

The Placement Effort Level options available in the Automatic Placement form are *Low*, *Medium*, and *High*.



This page does not contain notes.

Quiz: Parameters Tab in the Placement Planning Form



What are the Parameters tab available in the Placement Planning form?
(Select all the correct choices.)

- A. Tap cell
- B. Device
- C. Area
- D. Row
- E. Boundary Cell
- F. All of the above

Answers: A, D, and E

The Parameters tab available in the Placement Planning form are *Tap Cell*, *Row*, and *Boundary Cell*.



This page does not contain notes.



Module 3

Binder and Extractor

cādence®

This page does not contain notes.

Module Objectives

In this module, you

- Analyze how the Virtuoso® Layout-EXL Binder and Extractor work
- Debug connectivity problems
- Define advanced Series/Parallel mapping
- Display the connectivity information using the Annotation Browser Assistant and the Navigator Assistant



This page does not contain notes.



Submodule 3-1

XL Compliance

cādence®

This page does not contain notes.

Submodule Objectives

In this submodule, you:

- Identify the features of Binder
- Evaluate the significance of Binder
- Analyze the principles of XL Compliance
 - Assess the methods for verifying XL Compliance
 - Create a plan for maintaining XL Compliance
 - Analyze the process of Manual Binding
 - Describe how to stay XL Compliant with Manual Binding
 - Apply the Define Device Correspondence (DDC) command
 - Explain Cross-Probing and XL Compliance After Manual Binding



This page does not contain notes.

What Is the Binder and Why Is It Important?



- The **Binder** is central to the schematic-driven layout in EXL.
- It manages the mapping between schematic and layout instances.

It is essential that this mapping is correct, as logical connectivity from the schematic instance is added to layout instance terminals and propagated by extraction.

When working in the Layout EXL connectivity-aware environment, it is extremely important to establish and maintain the correct correspondence between the schematic and layout instances.

- The Layout EXL Binder makes and manages this correspondence.
- It allows the extractor to propagate the correct connectivity through the design.

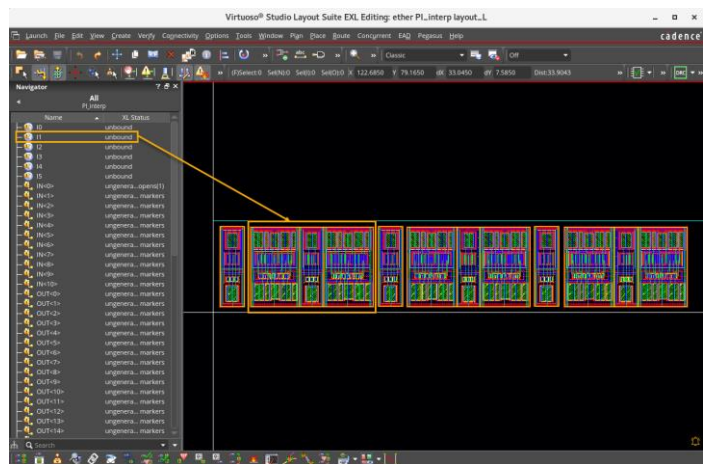
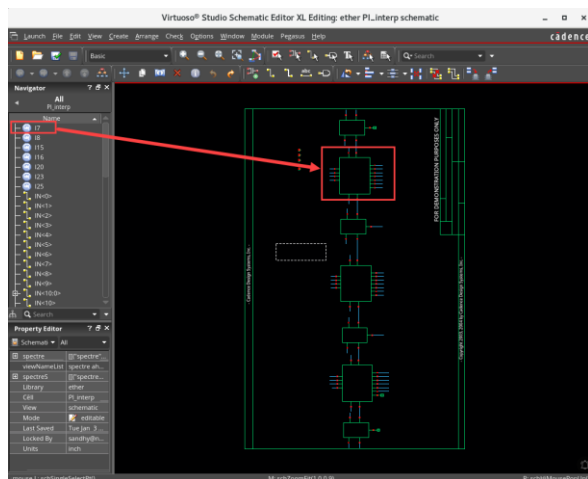


When working in the Layout EXL connectivity-aware environment, it is extremely important to establish and maintain the correct correspondence between the schematic and layout instances. The Layout EXL Binder makes and manages this correspondence, allowing the extractor to propagate the correct connectivity through the design.

Layout EXL mechanism used to determine and maintain device correspondence between a schematic and layout cellview pair.

Binder: Example

Schematic Instance is bound to the corresponding Layout Instance, as shown by the cross selection between Navigators and Canvases.



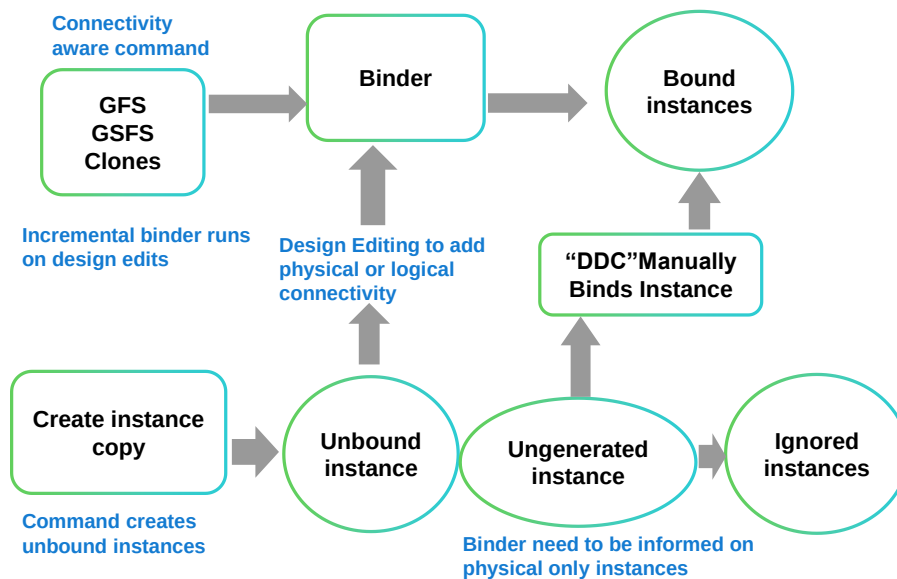
Logical connectivity from the schematic is added to the terminals of the layout instance.

You see the Binder example here.

Schematic Instance is bound to the corresponding Layout Instance as shown by the cross selection between Navigators and Canvases.

Logical connectivity from the schematic is added to the terminals of the layout instance.

Flowchart: Working with the Layout XL Binder



63 © Cadence Design Systems, Inc. All rights reserved.



When using connectivity-aware commands such as Generate All From Source (GFS), Pick From Schematic (PFS), and Clone commands, fully bound instances will be created.

When using non-connectivity aware Virtuoso commands such as Create Instance and Copy, unbound instances will be created.

You can edit the design to add physical or logical connectivity, for example, by routing, and when there is sufficient connectivity information to consider the instance unique, the Incremental Binder runs on design edits and automatically binds it to its schematic counterpart and creates fully bound instances.

In case of Ungenerated Instances, you can manually bind the instances using the Define Device Correspondence (DDC) command.

Binder needs to be informed on physical-only instances by adding the *Ignore* property to make them as Ignored Instances.

Overview: XL Compliance



What is XL Compliance?

- Pins and Instances are correctly bound between schematic and layout.
- Correct logical (from schematic) connectivity propagated through the layout.
- *XL Status* and *Check Against Source* report XL Compliance.

Why is XL Compliance important?

- Advanced functionality in XL and EXL needs schematic-layout correspondence and logical connectivity to be correct to work correctly.

How can you stay XL Compliant?

- Work in Virtuoso LS XL/EXL to take advantage of the Binder and Extractor.
- Use the XL generation commands where possible: *GFS/GSFS/Clone*.

What are the sources of Non-XL Compliance?

- Working outside of Virtuoso LS XL/EXL or using non-connectivity aware commands: *Create Instance*, *Mosaics*, or *Copy*.

64

© Cadence Design Systems, Inc. All rights reserved.



In the simplest terms, XL Compliance is having pins and instances correctly bound between the schematic and layout. This is important to allow the correct logical (from schematic) connectivity to be propagated through the layout.

Unbound and ungenerated instances or terminals are the most obvious examples of non XL Compliance that need to be investigated.

An unbound instance or terminal is one that exists in the layout but does not have a corresponding instance or terminal in schematic. This may be because it does not exist or the correspondence has not been made.

An ungenerated instance or terminal is one that exists in the schematic but does not have a corresponding instance or terminal in the layout. Again, this may be because it does not exist or the correspondence has not been made.

Layout Binding: Automatic Binding Versus On-Demand Binding



Automatic Binding:

When Layout XL is started and when editing the layout in XL, the binder is invoked.

Initial Binding

- When a layout is opened in XL, the Binder is invoked.
- If the layout has been previously opened in XL, the Binder respects the existing bindings. Otherwise, it attempts to bind only instances with matching names between the schematic and layout.

Incremental Binding

- When editing a layout in XL, the Binder is running.
- Components created using the connectivity-aware commands are automatically bound to the schematic.
- Any unbound instance is bound automatically if sufficient connectivity is added.
 - For example, by routing – to make it a unique match to a schematic instance with the same master.



On-Demand Binding

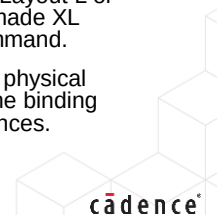
Use the Define Device Correspondence (DDC) and Update Binding commands to stay XL Compliant as required.

Manual Binding

- You can make or change a binding manually at any time in Layout XL by using Define Device Correspondence (DDC).
- These user-defined bindings are respected in subsequent XL sessions and are highlighted in the DDC form and in the Layout Navigator.

Update Binding

- LVS-clean legacy designs completed in Layout L or designs with incorrect bindings can be made XL compliant using the Update Binding command.
- The Update Binding command uses the physical connectivity in the layout to determine the binding between the layout and schematic instances.



Layout XL Binder's operation is often invisible to the user. The XL Binder functionalities are:

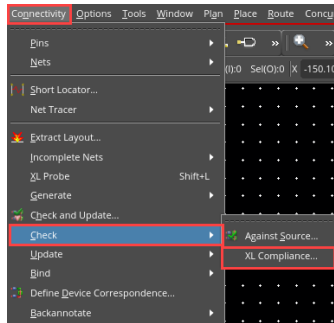
- **Automatic Binding**
 - Initial Binding
 - Incremental Binding
- **On-Demand Binding**
 - Manual Binding
 - Update Binding

Checking XL Compliance: XL Compliance Report in CIW



In the layout canvas, choose **Connectivity – Check – XL Compliance** to get the XL Compliance status in CIW during an XL session.

When **Connectivity – Check – XL Compliance** is run, an on-demand check of the binding status and design compliance is obtained during an XL session.



Virtuoso® Studio IC25.1 - Log: /home/sandhy/CDS.log.1

cadence

To get an accurate value, set the 'checkExtractLayout' environment variable to true.

Summary	Bound	Unbound	Ungenerated	Direction Differences	%Compliance
Terminals	33	0	2	22	57
Instances	0	6	7		0
Nets	33	0	11		75
Total	66	6	20	22	44

An on-demand check of the binding status and design compliance during session

An on-demand check of the binding status and design compliance during an XL session.

When **Connectivity – Check – XL Compliance** is run, an on-demand check of the binding status and design compliance is obtained during an XL session.

Checking XL Compliance: Binder Report in CIW



Open the layout in EXL by choosing **Launch – Layout EXL** from the schematic or by choosing **File – Open** from the CIW or by **right-clicking** and opening the layout in EXL from the Library Manager.

When the layout is opened in Virtuoso LS EXL, Binder Report is written in CIW.

The screenshot shows the Virtuoso Studio IC25.1 interface. On the left, the 'Launch' menu is open, with 'Layout EXL' highlighted. The main window displays the 'PI_interp/schematic' layout. A table in the CIW shows the following data:

	bound	unbound	ungenerated
terminals	33	0	2
nets	33	0	11
instances	0	6	7

Below the table, a warning message is displayed:
WARNING (BND-2005): Layout instances and terminals do not match the source.
Use 'XL Status' in the Navigator or run 'Connectivity - Check - Against Source' for more details.
If the bindings are correct, run the 'Update from Source' commands, otherwise run 'Connectivity - Update - Binding'.
An arrow points from the text 'Need Investigation' to the warning message.

Unbound or ungenerated instances need investigation.



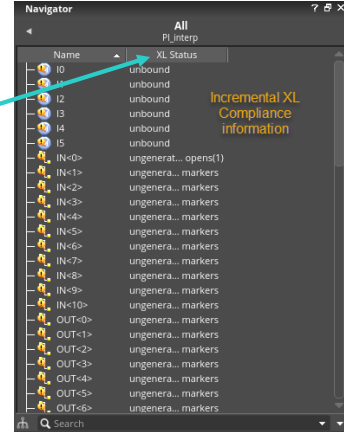
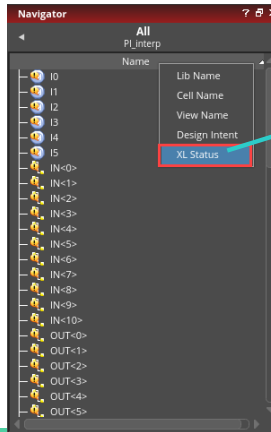
When the layout is opened in Virtuoso LS EXL, Binder Report is written in CIW. Unbound or ungenerated instances need investigation.

Checking XL Compliance: Layout Navigator XL Status Column



In the layout canvas in the Navigator assistant, **right-click** in the menu banner to activate the **XL Status** Column.

Layout Navigator XL Status Column gives Incremental XL Compliance information.



Incremental XL Compliance information



Layout Navigator XL Status Column gives Incremental XL Compliance information.

Staying XL Compliant with the Generation Command: GFS



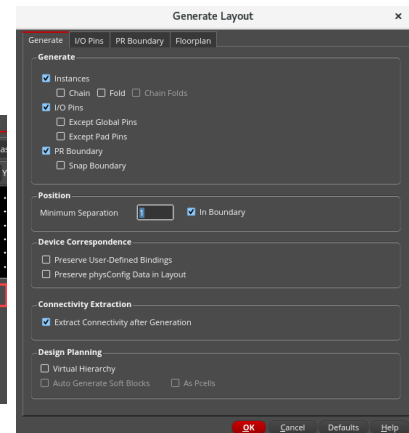
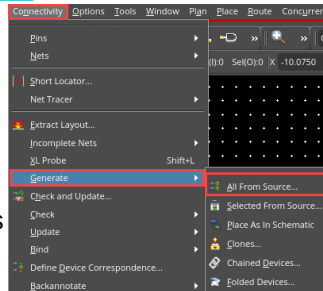
Connectivity – Generate – All From Source or Generate All From Source (GFS) icon at the *lower-left* corner of the layout canvas.



Generate All From Source will create layout instances for all schematic instances.

Advantages

- All instances XL Compliant
- Instances bound to their schematic counterparts and have the correct connectivity
- Cross-selection and probing between schematic and layout
- Instances positioned roughly as in the schematic



Note: **Create Instance** is not connectivity-aware and does not create instances bound to schematic.

This page does not contain notes.

Staying XL Compliant with the Generation Command: GSFS

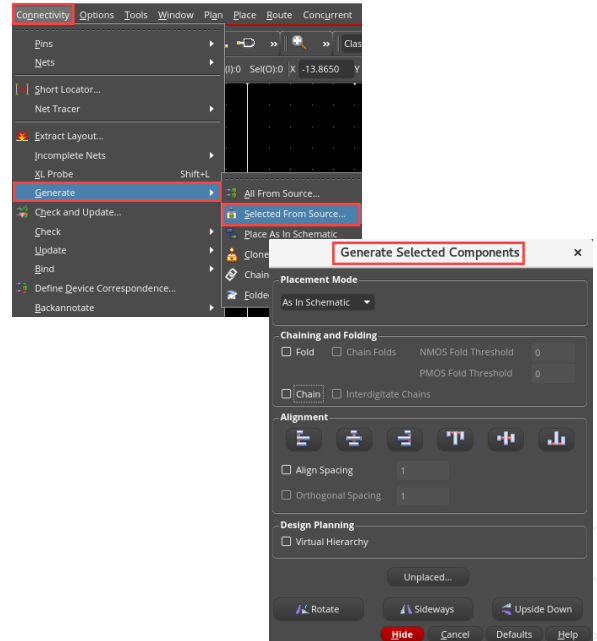
Generate Selected From Source allows incremental layout generation.



Advantages

- All instances XL Compliant
- Instances bound to their schematic counterparts and have the correct connectivity
- Cross selection and probing between schematic and layout
- Instances positioned roughly as in the schematic
- Have added control on placement
- Driven from the schematic or layout Navigator Assistant

Note: **Create Instance** is not connectivity-aware and does not create instances bound to schematic.



70 © Cadence Design Systems, Inc. All rights reserved.

To generate a layout that is XL Compliant by construction, it is recommended that you use the connectivity-aware layout generation commands as appropriate for your design situation.

Use this XL generation command where possible: *GSFS*.

You can display this form by choosing either the **Connectivity – Generate – Selected From Source** pull-down menu or by clicking the **Generate Selected From Source (GSFS)** icon at the *lower-left* corner of the layout canvas.

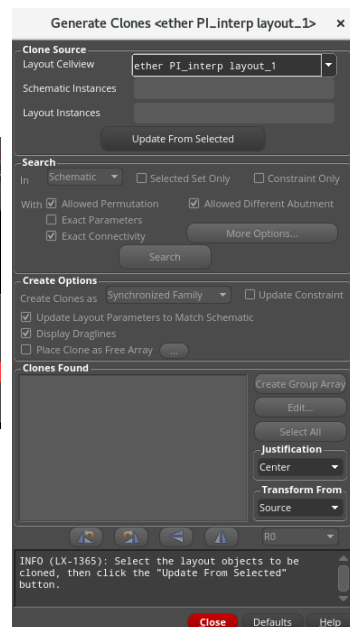
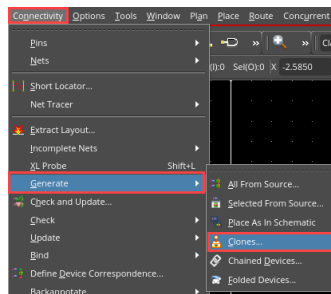
Staying XL Compliant with the Clone Command: Clones

Generate Clones is a connectivity-aware Copy command and creates XL Compliant instances.

Advantages

- Searches the schematic for matches and reuses existing placements
- Synchronous behavior for future edits

Note: **Copy** is not connectivity-aware and does not create instances bound to schematic.



To increase productivity, it is important to be able to take a subsection of the design that is already laid out and replicate it in the layout. The Generate Clones command allows you to do this in a connectivity-aware, XL Compliant way.

Use this XL generation command where possible: *Clones*.

You can display this form by choosing either the **Connectivity – Generate – Clones** pull-down menu or by clicking the **Generate Clones** icon at the *lower-left* corner of the layout canvas.



Submodule 3-2

Manual Binding

cādence®

This page does not contain notes.

Submodule Objectives

In this submodule, you

- Analyze the process of Manual Binding
- Describe how to stay XL Compliant with Manual Binding
- Apply the Define Device Correspondence (DDC) command
- Explain Cross-Probing and XL Compliance After Manual Binding



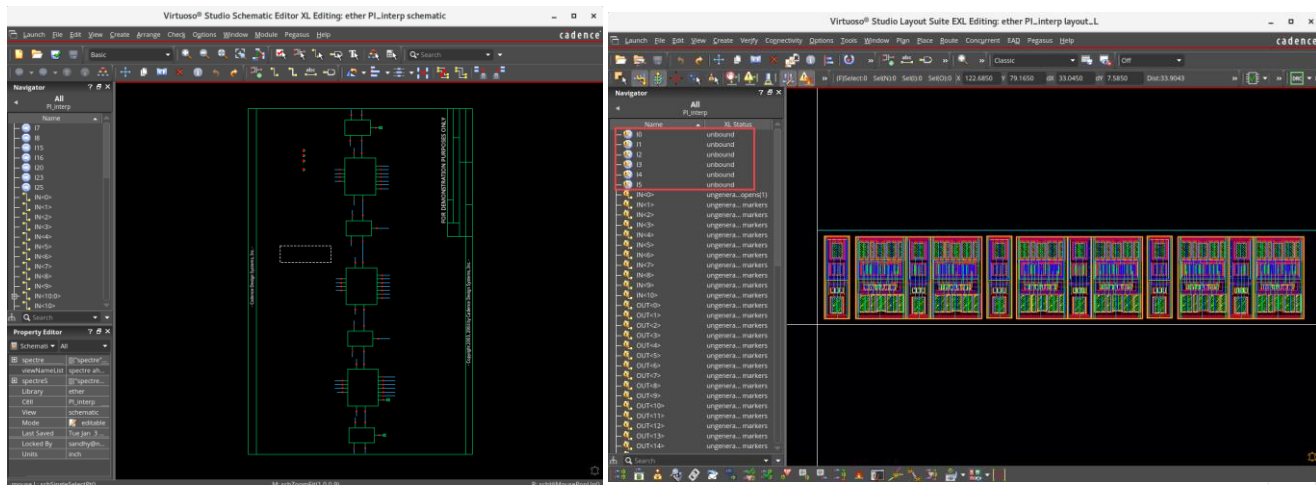
This page does not contain notes.

Staying XL Compliant with Manual Binding

Create Instance, Copy, and Manual Binding



- Layout designers prefer flexibility in using **Create Instance** or **Copy**.
- Supported flow but requires **Manual Binding** to stay **XL Compliant**.



74 © Cadence Design Systems, Inc. All rights reserved.



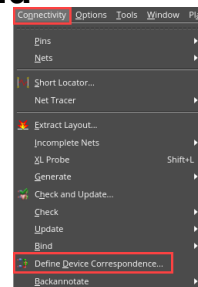
In some situations, you may use the Create Instance or Copy command when creating a layout. This is a supported flow in Layout EXL but not recommended. Therefore, making the generated layout XL Compliant requires additional effort. The unbound instances created by these commands require to be bound to their schematic counterpart. One of the ways you can achieve this is by applying Manual Binding using Define Device Correspondence (DDC).

You can display this form by choosing either the **Connectivity – Define Device Correspondence** pull-down menu or by clicking the **Define Device Correspondence (DDC)** icon at the *lower-right* corner of the layout canvas.

Running the Define Device Correspondence Command



Run Define Device Correspondence by choosing either the **Connectivity – Define Device Correspondence** menu command or by clicking the **Define Device Correspondence (DDC)** icon on the Layout EXL Toolbar.

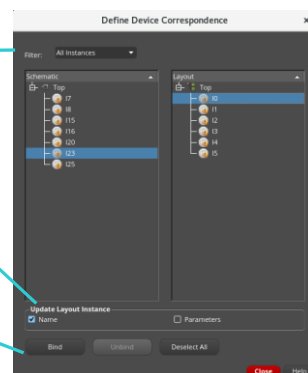


- Use the Define Device Correspondence form to create or update the binding between schematic and layout instances or terminals for which correspondence is defined.
- Device correspondence also propagates connectivity of the wiring connections between the schematic and layout.

Select **All Instances** in the **Filter** cyclic field to show also the already mapped ones.

Updates the layout instance name to match the corresponding schematic instance.

Creates a correspondence between the devices selected in the Schematic pane and the devices selected in the Layout pane. This button is enabled only if all the selected devices are unbound.



75 © Cadence Design Systems, Inc. All rights reserved.

cadence

You use the Define Device Correspondence form to create or update the binding between schematic and layout instances or terminals for which correspondence is defined.

Device correspondence also propagates connectivity of the wiring connections between the schematic and layout.

To define the device correspondence:

1. Choose the **Connectivity – Define Device Correspondence** menu entry or the **Define Device Correspondence** icon to display the form.
2. Choose an option from the Filter cyclic list: *All Instances*, *All Terminals*, *Unbound Instances* (the default), or *Unbound Terminals*.
3. Click an instance in either column. The instance is selected in the other column and in the schematic and layout windows.
4. Change the binding if desired.

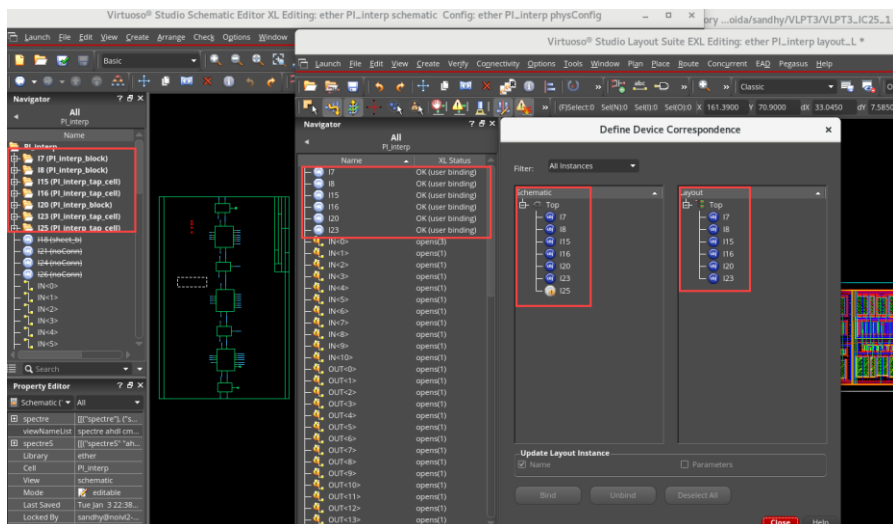
If there is no binding for the item you selected, the Bind button becomes available. Choose an item from the other column and click **Bind**.

If the item you selected is bound, the Bind button is grayed out but the Unbind button is available. If you want, you can unbind the item and bind it to a different instance.

See complete details on the use of this form by clicking the **Help** button in the *lower-right* corner of the form.

Example: Cross-Probing and XL Compliance After Manual Binding

- Schematic and Layout are with XL Compliance.
- The layout instance names match the schematic instance names.



76 © Cadence Design Systems, Inc. All rights reserved.

cadence

Cross-probing lets you select an object in the *layout* and the software subsequently selects and highlights that object in the *schematic*, and vice versa. Also, an object selected in the Navigator, Annotation Browser, or Constraint Manager will cross-select and highlight as well.

Because this is connectivity driven.



Submodule 3-3

Incremental Binding

cādence®

This page does not contain notes.

Submodule Objectives

In this submodule, you:

- Analyze the process of Incremental Binding
- Describe how to stay XL Compliant with Incremental Binding/Extraction
- Use the Copy Command
- Use the Create Wire Command
- Generate Selected From Source (GSFS) from the Navigator assistant
- Update the Components and Nets
- Explain Cross-Probing and XL Compliance After Incremental Binding

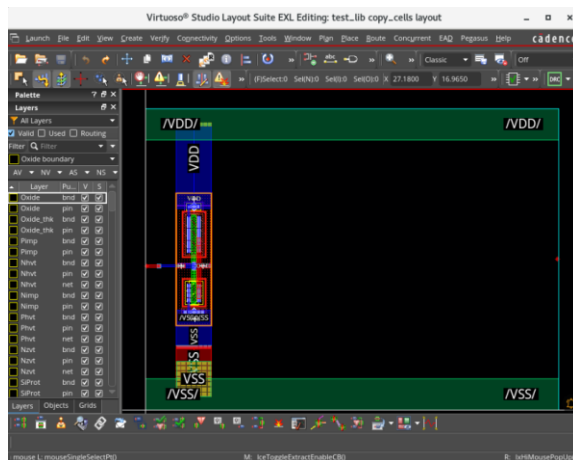
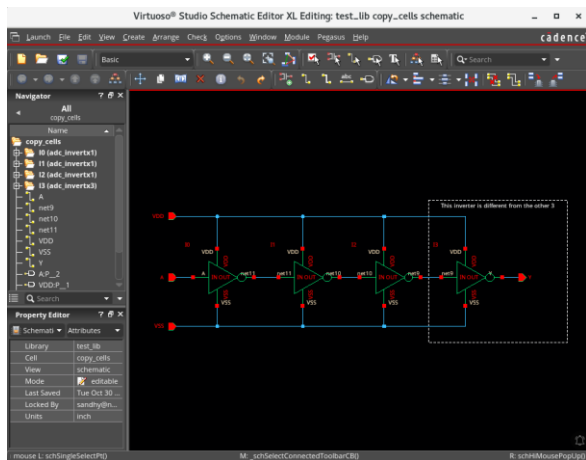


This page does not contain notes.

Staying XL Compliant with Incremental Binding Create Instance, Copy, and Incremental Binding/Extraction



- Layout designers prefer flexibility in using **Create Instance** or **Copy**.
- Supported flow but requires **Incremental Binding** to stay **XL Compliant**.



In some situations, you may use the Create Instance or Copy command when creating a layout. This is a supported flow in Layout EXL; therefore, making the generated layout XL-compliant requires additional effort. The unbound instances created by these commands are required to be bound to their schematic counterpart. One of the ways you can achieve this is by using *Automatic Incremental Binding* when connectivity is added to the layout.

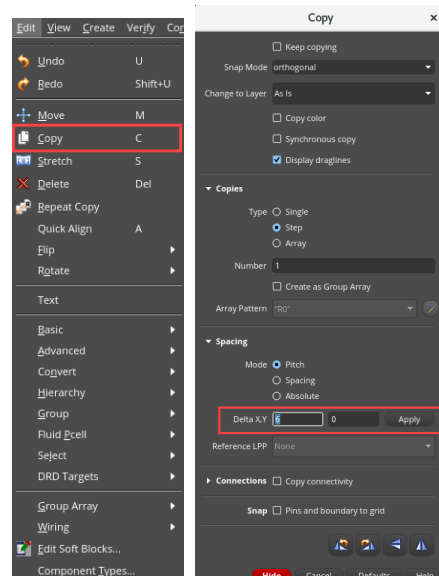
An alternative to *Manual Binding* is to add connectivity to the layout with routing. When there is sufficient connectivity information to consider the instance unique, the Incremental Binder automatically binds it to its schematic counterpart.

Using the Copy Command



Invoke the Copy form by choosing either the **Edit – Copy** menu command or by pressing the bindkey **C** or by clicking the **Copy** icon on the **Edit** toolbar.

Since the Copy command is used, the new copies will initially be unbound, and VLS-EXL will NOT know anything about them.



In the Copy form, in the **Delta X Y** field, enter **X=6**. Click the **Apply** button twice to create two copies of the inverter at an interval of **6 units** along the **X-axis**.

80 © Cadence Design Systems, Inc. All rights reserved.

cadence

Let us see how to invoke and use the *Copy* command.

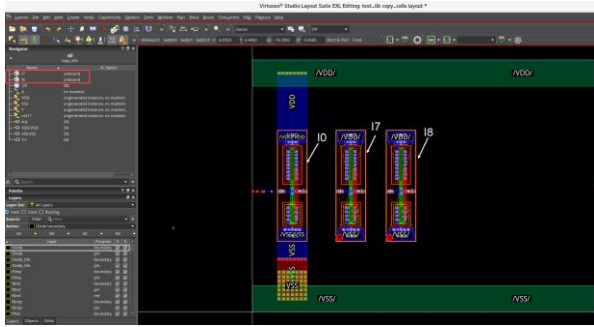
You can invoke the Copy form by choosing either the **Edit – Copy** menu command or by pressing the bindkey **C** or by clicking the **Copy** icon on the Edit toolbar.

Since the L-Level Copy command is used, the new copies will initially be unbound and Virtuoso Layout Suite EXL will not know anything about them.

In the Copy form, in the *Delta X Y* field, enter $X=6$.

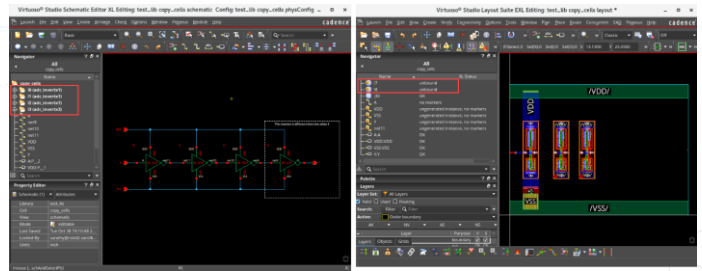
Click the **Apply** button twice to create two copies of the inverter at an interval of *6 units* along the *X-axis*.

Result of the Copy Command



Two copies of the inverter at an interval of **6 units** along the **X-axis** are created.

The instance names of the two copies of the layout do not reflect the names in the schematic.



Two copies of the inverter at an interval of 6 units along the X-axis are created.

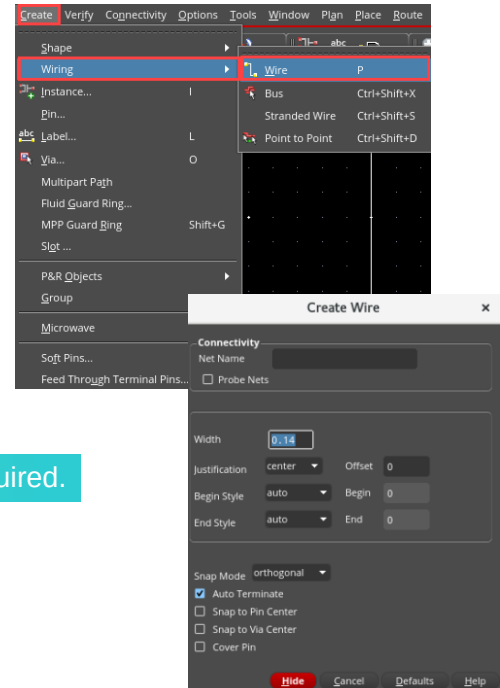
The instance names of the two copies of the layout do not reflect the names in the schematic.

Also the cross-probing between schematic/layout is not working as expected for these two copies of the layout because they are NOT bound yet.

Using the Create Wire Command



Invoke the Create Wire form by choosing either the **Create – Wiring – Wire** pull-down menu or pressing the bindkey **P** or clicking the **Create Wire** icon on the Create toolbar.



- As soon as the connection is completed, the Binder identifies the instances.
- The cross-probing between schematic/layout is now working.
- The instances are marked as **OK** in XL Status column.

Start interactive routing and complete the connection as required.

Let us see how to invoke and use the Create Wire command.

You can invoke the Create Wire form by choosing either the Create – Wiring – Wire pull-down menu or pressing the bindkey **P** or clicking the **Create Wire** icon on the Create toolbar.

Start interactive routing and complete the connection as required.

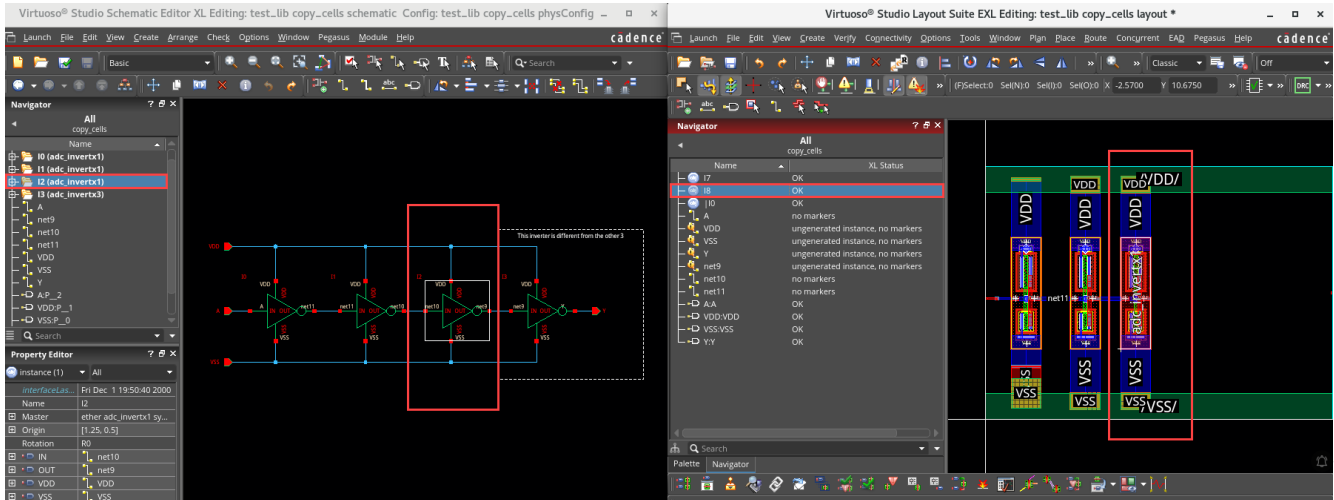
As soon as the connection is completed, the Binder identifies the instances.

The cross-probing between schematic/layout is now working.

The instances are marked as OK in the XL Status column.

Result of the Create Wire Command

The cross-probing between schematic/layout is now working as expected.



The instances are marked as OK in XL Status column.

As soon as the connection is completed, the cross-probing between schematic/layout is now working as expected.

The instances are marked as OK in the XL Status column.

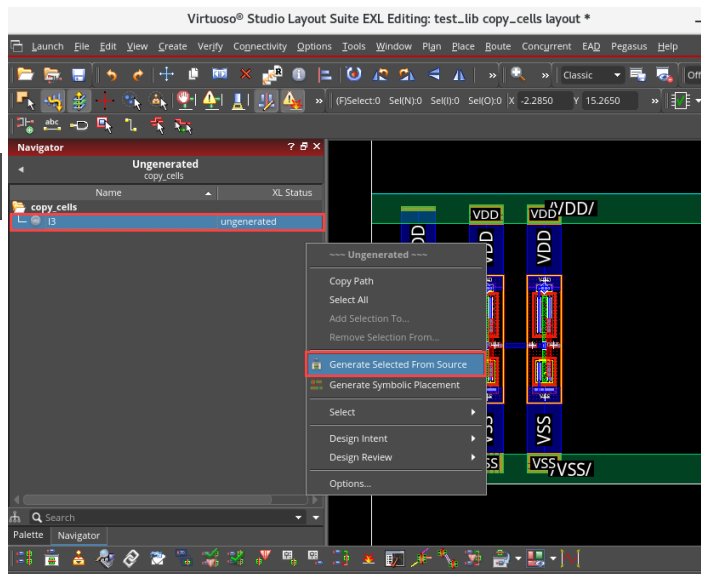
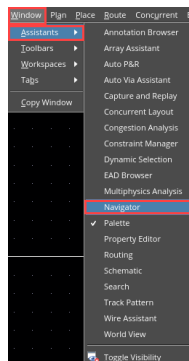
Generating Selected From Source (GSFS) from the Navigator



In the layout canvas, choose **Window – Assistants – Navigator** to activate the **Navigator** Assistant. In the Navigator, **right-click** on an instance and select **Generate Selected From Source (GSFS)** command option.



The device is selected and remains glued on your cursor until you place it in the layout.



84 © Cadence Design Systems, Inc. All rights reserved.



Let us see how to Generate Selected From Source from the Navigator Assistant.

In the layout canvas, choose **Window – Assistants – Navigator** to activate the Navigator assistant if it is not already active.

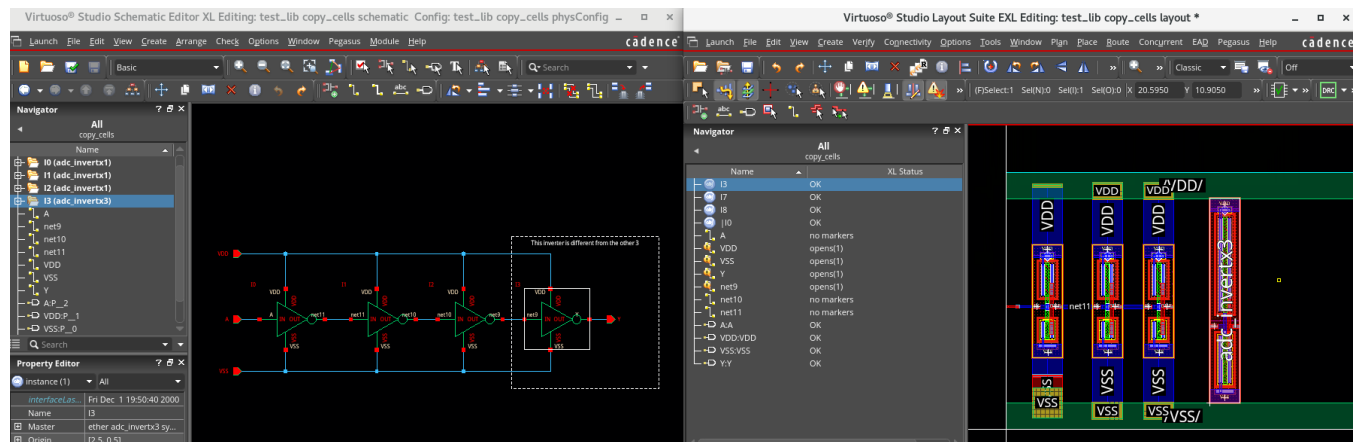
In the Navigator assistant, **right-click** on an instance and select the **Generate Selected From Source** command option.

When the GSFS command is invoked, the device is selected and remains glued on your cursor until you place it in the layout.

You can now place the device in the layout.

Result: GSFS Command

The instance I3 is placed in the layout from the source schematic view.



85 © Cadence Design Systems, Inc. All rights reserved.

cadence

The instance I3 is placed in the layout from the source schematic view.

The layout placement is now complete.

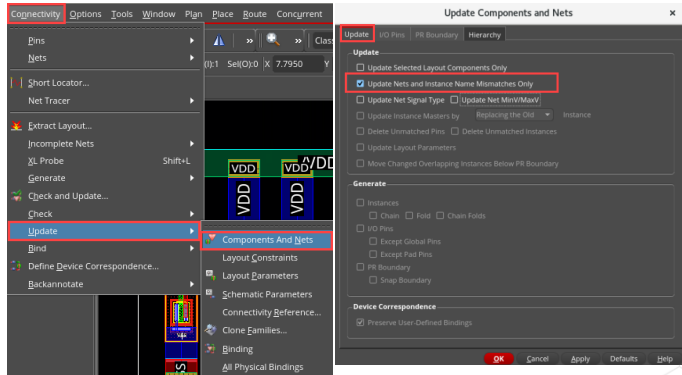
All the instances are in the layout view and they are all *bound*, as you can easily verify cross-probing them.

Updating Components and Nets



Invoke the Update Components and Nets form by choosing either the **Connectivity – Update – Components And Nets** menu entry or clicking the **Update Components and Nets (UCN)** icon on the Layout EXL toolbar.

- UCN makes the naming consistent between schematic/layout.
- In the UCN form, activate the **Update Nets and Instance Name Mismatches Only** checkbox.
- UCN changes the name of the layout instances to match the ones used in the schematic.



86 © Cadence Design Systems, Inc. All rights reserved.

cadence

Let us see how to Update the Components and Nets between schematic/layout.

You can invoke the Update Components and Nets form by choosing either the **Connectivity – Update – Components And Nets** menu entry or by clicking the **Update Components and Nets (UCN)** icon on the Layout EXL toolbar.

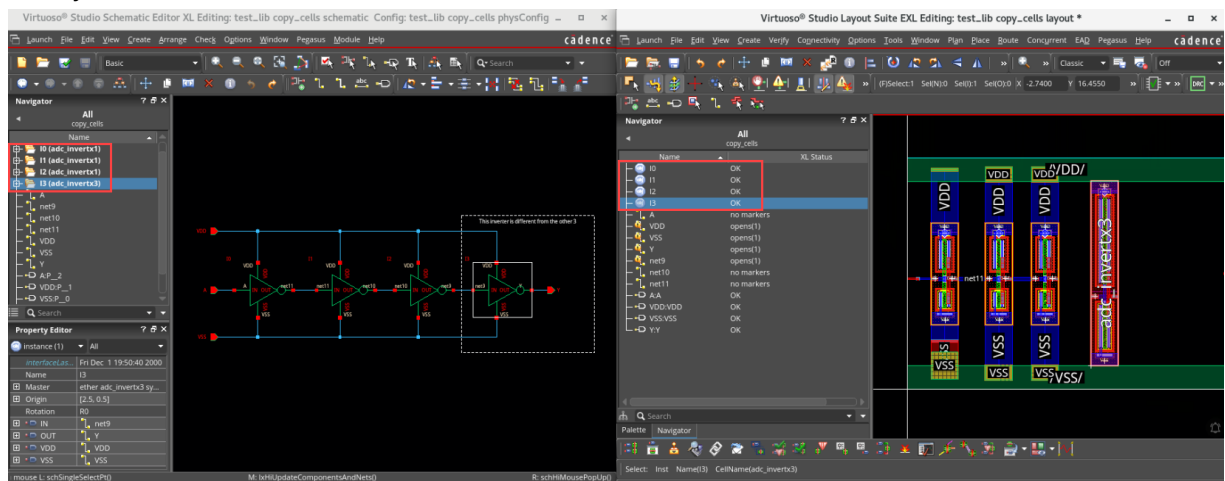
You can make the naming consistent between schematic/layout using the Update Components and Nets command.

In the Update Components and Nets form, activate the **Update Nets and Instance Name Mismatches Only** checkbox.

UCN changes the name of the layout instances to match the ones used in the schematic.

Example: Cross-Probing and XL Compliance After Incremental Binding

- Schematic and Layout are with XL Compliance.
- The layout instance names match with that of the schematic instance names.



Cross-probing lets you select an object in the *layout* and the software subsequently selects and highlights that object in the *schematic*, and vice versa. Also, an object selected in the *Navigator*, *Annotation Browser*, or *Constraint Manager* will cross-select and highlight as well.



Submodule 3-4

Update Binding

cādence®

This page does not contain notes.

Submodule Objectives

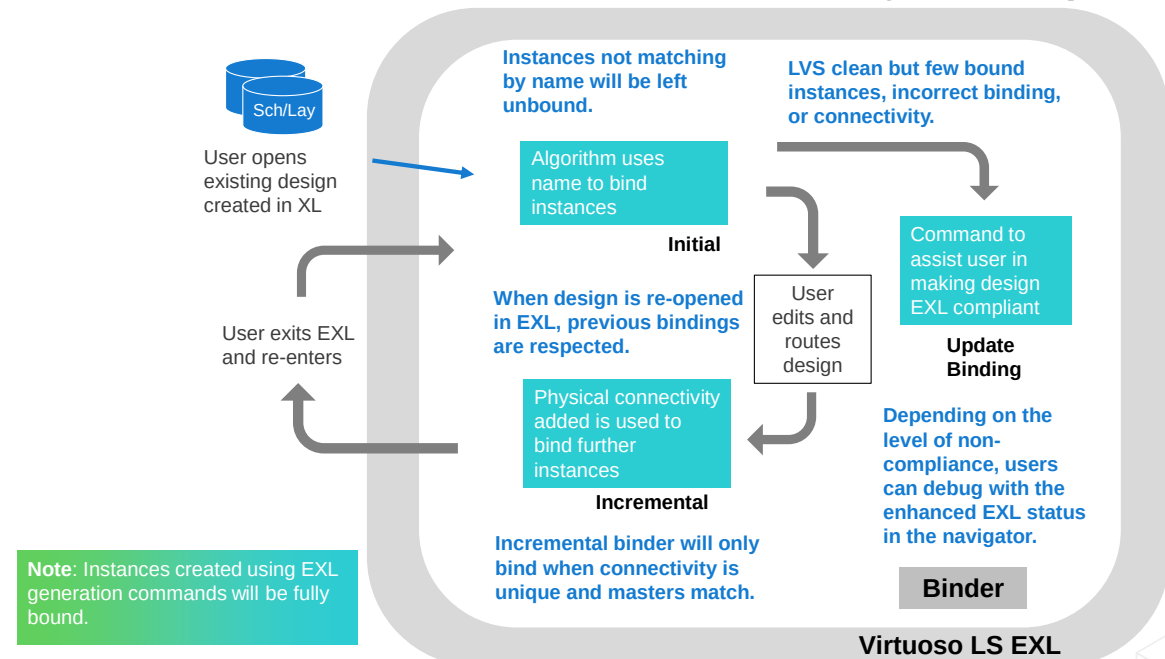
In this submodule, you

- Analyze the process of Update Binding
- Describe how to stay XL Compliant with Update Binding
- Run the Update Binding command



This page does not contain notes.

Flowchart: Virtuoso LS EXL Binder: Increasing XL Compliance



Let us see about the Virtuoso Layout Suite XL Binder and how it helps to increase the XL Compliance of our design.

Layout XL Binder's operation is often invisible to the user.

When a layout is opened in EXL, the Binder is invoked.

If the layout has been previously opened in EXL, the Binder respects the existing bindings. Else, it attempts to bind only instances with matching names between the schematic and layout.

When editing a layout in EXL, the Binder is running.

Components created using the connectivity-aware commands are automatically bound to the schematic.

Any unbound instance is bound automatically if sufficient connectivity is added – for example, by routing – to make it a unique match to a schematic instance with the same master.

LVS-clean legacy designs completed in Layout L or designs with incorrect bindings can be made XL compliant using the Update Binding command.

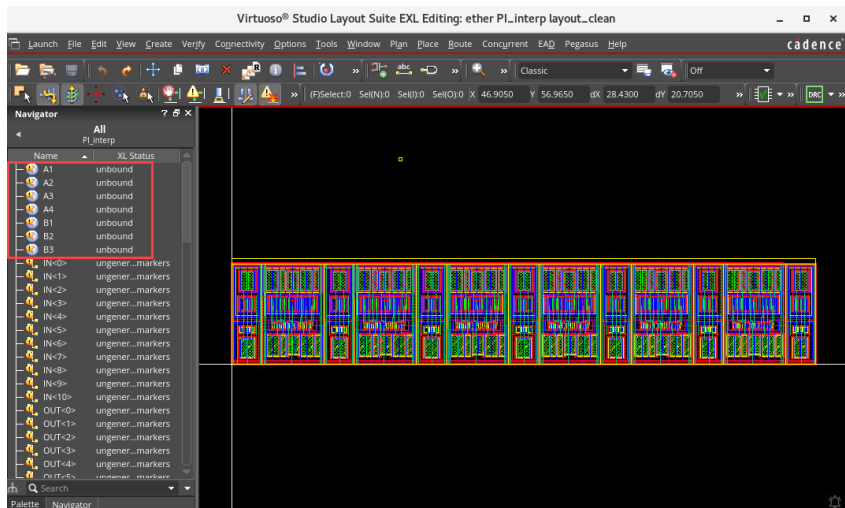
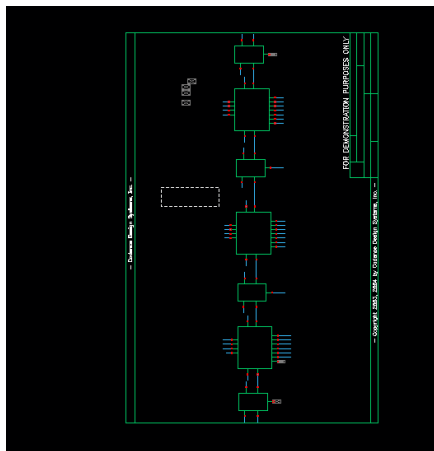
The Update Binding command uses the physical connectivity in the layout to determine the binding between the layout and schematic instances.

Note: Instances created using XL generation commands will be fully bound.

Staying XL Compliant with Update Binding Create Instance, Copy, and Update Binding



- Layout designers prefer flexibility in using **Create Instance** or **Copy**.
- Supported flow but requires **Update Binding** to stay **XL Compliant**.



91 © Cadence Design Systems, Inc. All rights reserved.

cadence

In some situations, you might prefer to use the Create Instance or Copy commands when creating a layout. This is a supported flow in Layout XL but not recommended because neither of these commands is connectivity aware. Therefore, making the generated layout EXL Compliant requires additional effort. The unbound instances created by these commands require to be bound to their schematic counterpart. One of the ways you can achieve this is by using the Update Binding command.

Update Binding uses the physical connectivity in the layout, such as routing, to trace connectivity through the layout and make bindings based on that. This means it is important that the design is LVS clean. Incorrect physical connectivity can very likely prevent Update Binding from making the layout XL Compliant.

You can display this form by choosing either the **Connectivity – Update – Binding** pull-down menu or clicking the **Update Binding** icon on the Layout EXL toolbar.

Staying XL Compliant with Update Binding Use of Mosaics and Update Binding



Mosaics are commonly used in Virtuoso LS L but can cause issues in XL.

A mosaic has only a single set of *instTerms*, so all elements in the mosaic must have the same connectivity to be XL compliant.

- Mosaics whose elements do not have the same connectivity must be flattened to become XL Compliant.
- Binder and Extractor support for mosaics has been added.
- Update Binding command can bind correctly.
- Extractor can extract the connectivity.
- Check and Update functions are being enhanced to work fully with mosaics.



When you use non-connectivity aware commands, such as *Mosaics*, it may cause your layout to lose its XL Compliance.

Becoming XL Compliant with Update Binding

Reuse of LVS-Clean Layouts and Update Binding



I have a layout that is LVS clean. Can I use it in Virtuoso LS XL?

Productivity is increasingly important. Correct connectivity is needed to take advantage of the advanced functionality in LS XL/EXL.

Reusing LVS-clean layouts created outside of XL is vital.

- Previously, it was difficult and error prone to make a layout XL compliant, negating the LS XL productivity advantage.

Design issues prevented the successful import of layouts into LS XL.

- Hierarchy mismatches between the schematic and the layout.
- Route Cells and Mosaics.
- Missing permute rules and Configure Physical Hierarchy bindings.
- Missing instance pins and ambiguous circuits.

Update Binding is introduced to make the designs **XL Compliant**.

93 © Cadence Design Systems, Inc. All rights reserved.



Layout productivity and design turn-around-time is increasingly important. To use the productivity enhancing functionality in Virtuoso LS , the layout must be XL Compliant.

Legacy designs or those completed outside of Layout XL can be LVS clean but may not be XL Compliant. Previously, it was difficult and error prone to make such a layout XL Compliant, which would negate the LS XL productivity advantage. You can use the Update Binding feature to make this process quicker and easier.

Update Binding uses the physical connectivity in the layout, such as routing, to trace connectivity through the layout and make bindings based on that. This means it is important that the design is LVS clean. Incorrect physical connectivity can very likely prevent Update Binding from making the layout XL Compliant.

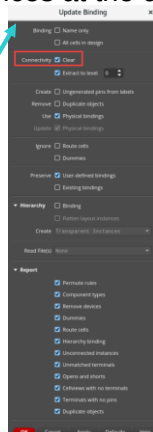
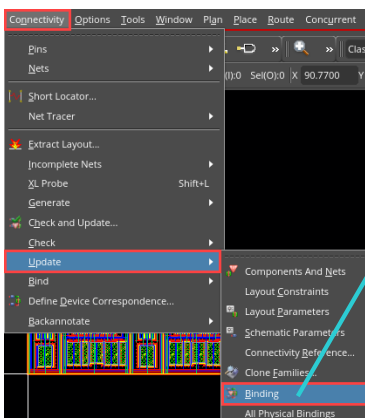
Running the Update Binding Command



Invoke the Update Binding form by choosing either the **Connectivity – Update – Binding** pull-down menu or by clicking the **Update Binding** icon on the Layout EXL toolbar.



Use the Update Binding command to increase Layout EXL Compliance by improving the binding between the schematic and the layout instances at the current level or within the hierarchy.



In the Update Binding form, select **Clear Connectivity** & set **Extract to level 0**.

94 © Cadence Design Systems, Inc. All rights reserved.

cadence

Use the Update Binding command to increase Layout EXL Compliance by improving the binding between the schematic and the layout instances at the current level or within the hierarchy.

If your design is largely LVS clean but the bindings are incorrect, you can clear the logical connectivity and re-extract the design.

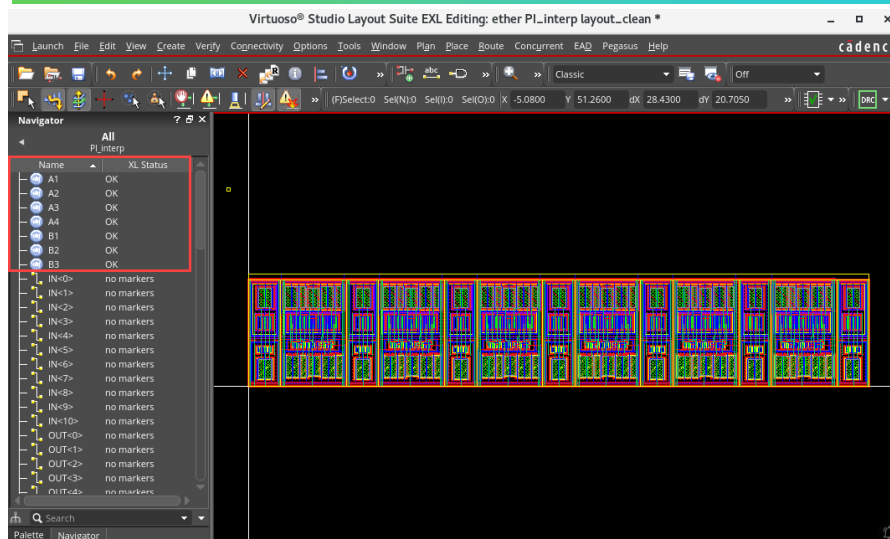
If you are sure that any user-defined bindings you created using the Define Device Correspondence form are correct, you can choose to preserve existing user-defined bindings. Alternatively, you can bind the instances based on a binding file.

You can invoke the Update Binding form by choosing either the **Connectivity – Update – Binding** pull-down menu or by clicking the **Update Binding** icon on the Layout EXL toolbar.

In the Update Binding form, select the **Clear Connectivity** option, and set the *Extract to level* field to 0, and run the Update Binding command.

Update Binding Result: Design Is Fully Bound

All the instances, terminals, and nets are bound.
This can be confirmed by checking the **XL Status** column in the Navigator.



95 © Cadence Design Systems, Inc. All rights reserved.

cadence

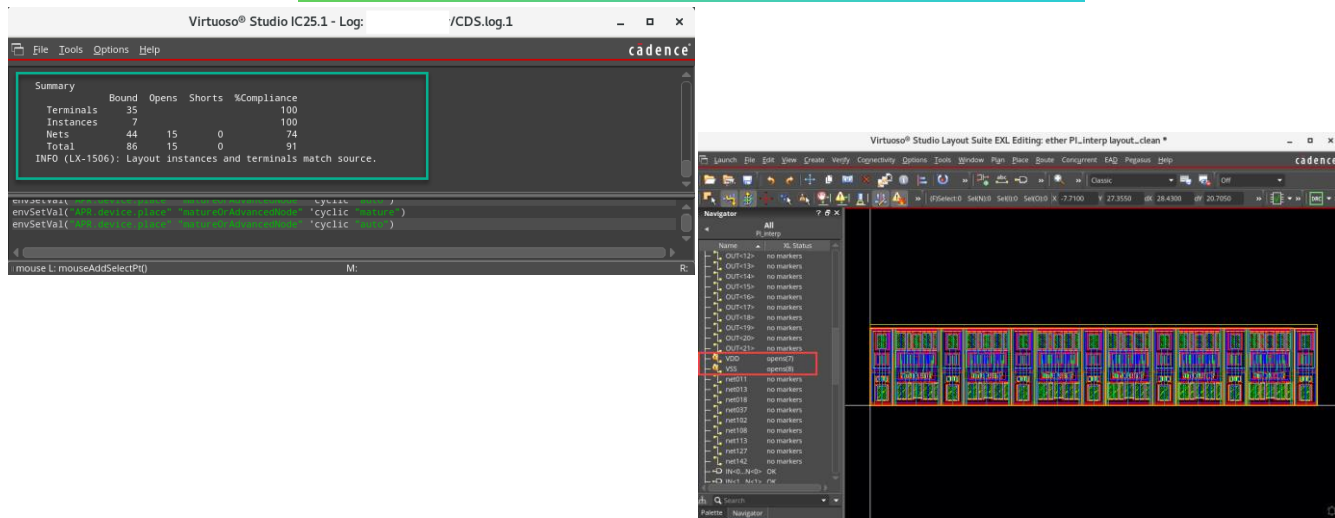
We see the Update Binding run result here and the design is fully bound.

We can see that all the instances, terminals, and nets are *bound*.

This can be confirmed, by checking the XL Status column in the Navigator assistant.

Update Binding Result: Design Is Fully Bound But with Opens

CIW reports **15 Opens** in the layout that need to be investigated.



The opens are on the **VDD** and **VSS** nets as seen in the Navigator.

We have seen that after the Update Binding runs, the design is fully bound, but there are some opens in the design.

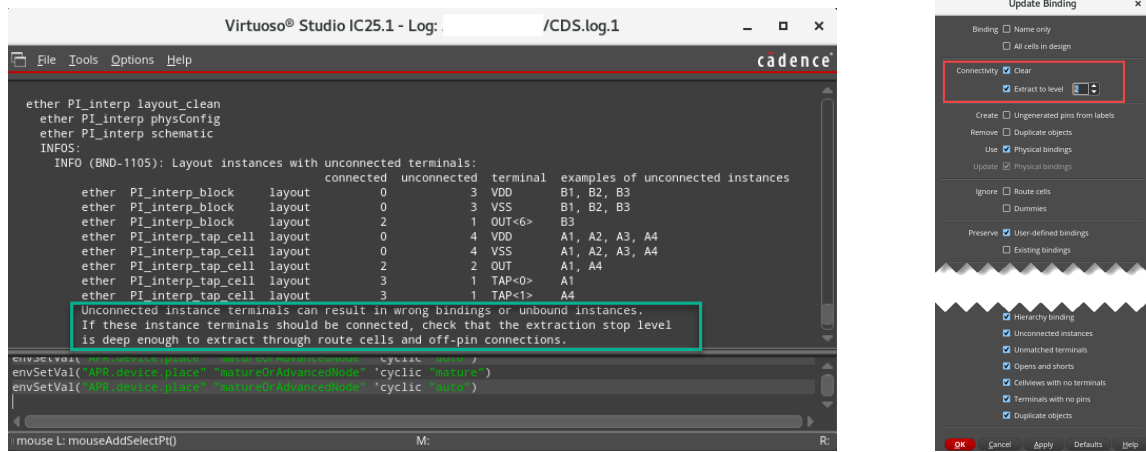
When **Update Binding** runs, it reports the binding status in the CIW window.

CIW window reports *16 Opens* in the layout that needs to be investigated.

The opens are on the *VDD* and *VSS* nets as seen in the Navigator assistant.

Update Binding Result: Increasing Extraction Level

Update Binding issues *INFO* messages to assist you with suggested actions to make the design XL Compliant.



Run **Update Binding** with **Extract to level** field to 2.

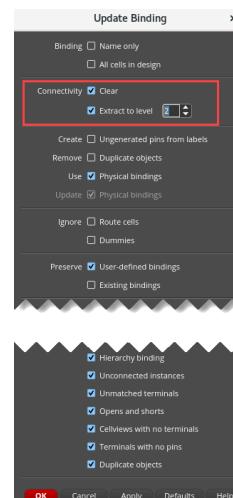
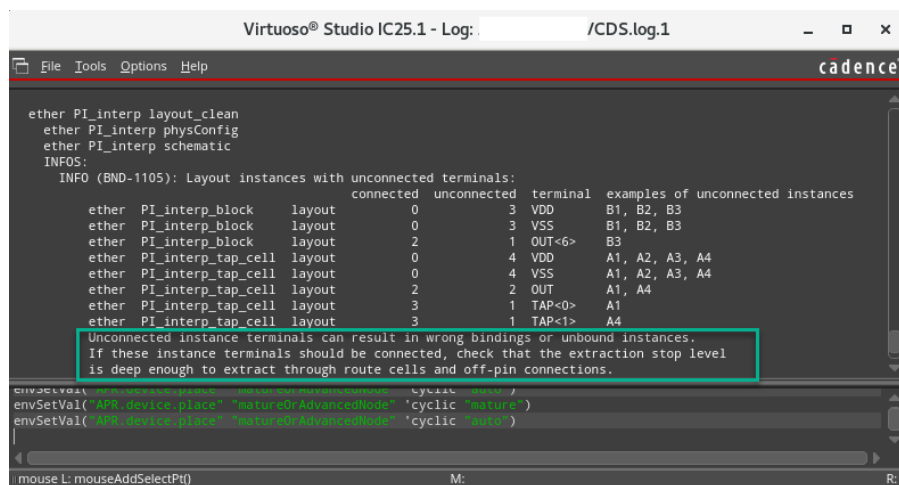
Let us see how to remove the opens by increasing the Extraction Level when running the Update Binding command.

Update Binding issues *INFO* messages to assist you with suggested actions to make the design XL Compliant.

Run Update Binding with *Extract to level* field to 2.

Update Binding Result: Design Is Fully Bound with No Opens

The layout is fully bound with **NO Opens and Shorts** and therefore, fully XL Compliant.



You can verify the **Update Binding Summary Report** in CIW.

After Update Binding completes, the layout is fully bound with *NO Opens* and *Shorts* and therefore, fully XL Compliant.

You can verify the same in the Update Binding Summary Report in the CIW window.

You can see that the opens on the *VDD* and *VSS* nets are removed after Update Binding by checking the XL Status in the Navigator assistant.

Update Binding: Things to Consider



During **Update Binding**, any layout-only instances (via cells, filler cells) should have an ignore property set on them so that EXL does not consider them for binding.

Update Binding warns of the following situations, which need to be resolved to get XL Compliant:

- **Tech File stopLayers**
 - Increasing extraction depth > 0 means that the stopLayers have to be correctly set in the technology file for transistors, resistors, and so on, so that the extractor does not create shorts.
- **Missing permuteRules**
 - In XL, devices are often copied and flipped so that their terminals (s/d or n/p) are connected interchangeably. In EXL, a permuteRule needs to be defined so that we know this is the case.
- **Missing Component Types**
 - Component types are used to identify devices, especially MOS devices, so that we can split multifingers to get the correct connectivity.
- **Mismatched terminals between schematic and layout instances**
 - Update Binding will report connectivity differences or may not bind if the terminals do not match between the schematic and the layout. Terminals to ignore or be mapped should be defined in LAM.

Underlying the Configure Physical Hierarchy functionality is a Layout EXL Schematic in IC 25.1. It comprises two new data formats, namely, LAM Files and Configuration Views.

For schematic and layout library data, there is a *library and attributes mapping (LAM)* file to store the symbol properties that drive layout generation, component type information for a given library, and library and cell-level mapping information. This file is called *cph.lam* and is stored in the following location:

library/.cadence/dfII/cph/lam/cph.lam

There is no graphical user interface to help you create or edit a LAM file. For more information on the LAM file syntax, see *Library and Attributes Mapping File Syntax* in *Virtuoso Layout Suite EXL: Connectivity Driven Editing, Product Version IC25.1*.

Update Binding: Things to Consider (continued)

Multi-Fingered Devices

- Multiple schematic instances implemented as a single multifingered device can cause some issues for Update Binding when attempting to put the correct connectivity on the individual fingers.

Flat Devices

- Flat devices are not supported. A cell should be created and then manually bound using DDC.



This page does not contain notes.



Submodule 3-5

Complex Binding

cādence®

This page does not contain notes.

Submodule Objectives

In this submodule, you

- Define Complex Binding
- Examine Complex Binding with lxCombination
- Generate All From Source with lxCombination
- Use the Annotation Browser to display the flight lines
- Run Check Against Source
- Create an Instance
- Run Define Device Correspondence
- Examine Cross-Probing and XL Compliance After Complex Binding



This page does not contain notes.

Complex Binding: *lxCombination*



You can use ***lxCombination*** to split a single resistor into multiple smaller devices.

- You can achieve maximum density in the layout by placing devices in spaces otherwise not utilized.
- This lets you approximate the resistance value defined in the schematic without violating the manufacturing grid.



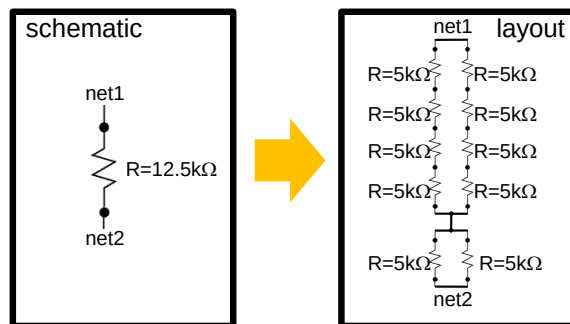
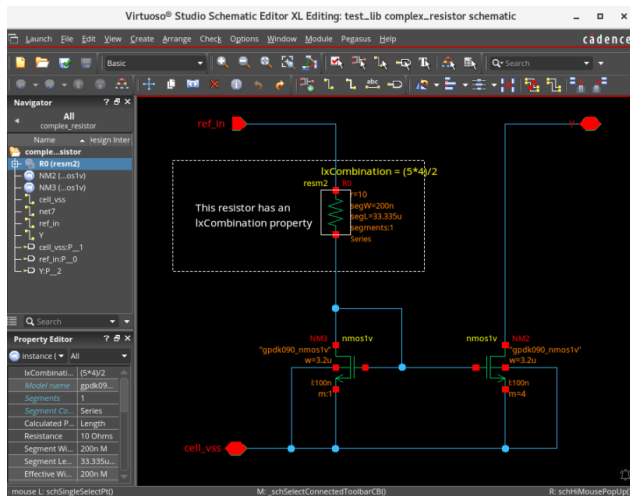
You typically use *lxCombination* to split a single resistor into multiple smaller devices to achieve maximum density in the layout by placing devices in spaces otherwise not utilized. This lets you approximate the resistance value defined in the schematic without violating the manufacturing grid.

The *lxCombination* property can be defined using the following operators:

- + and * are series connections.
- + can be followed by an expression; * must be followed by an integer.
- / is a parallel connection and division must be followed by an integer.
- Parentheses can be used to group expressions.

Complex Binding: lxCombination (Example)

- If the *lxCombination* property is defined on the schematic symbol, the *r* parameter is ignored by the Generate All From Source (GFS) and Update Components And Nets (UCN) commands.
- The layout instances will have the correct parameters based on the *lxCombination* factor setting.



$$lxCombination = (5k * 4)/2 + 5k/2 = 20k/2 + 2.5k = 10k + 2.5k = 12.5k$$

If the *lxCombination* property is defined on the schematic symbol, then the *r* parameter is ignored by the Generate All From Source (GFS) and Update Components And Nets (UCN) commands. The layout instances will have the correct parameters based on the *lxCombination* factor setting.

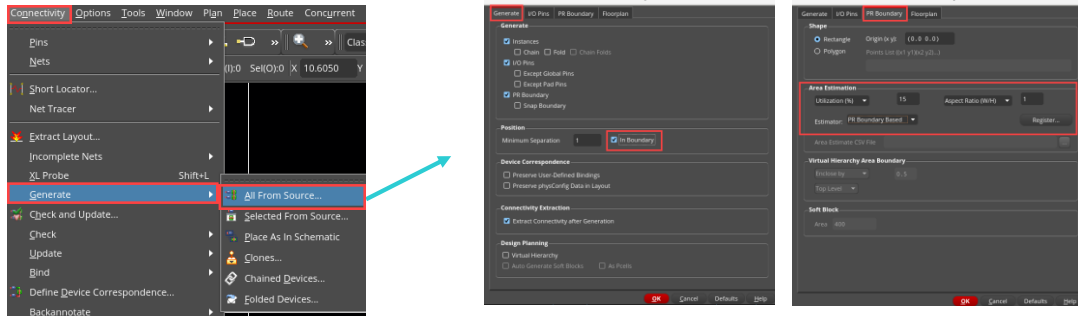
Note: You can also use *lxCombination* to define the topology for a network of capacitors. However, in this case, the property describes only the topology of the network. The values entered do not equate to the total capacitance of the network.

Generating All From Source



Generate all from the source by choosing either the **Connectivity – Generate – All From Source** menu command or clicking the **Generate All From Source (GFS)** icon. The Generate Layout form appears.

Use the GFS command to populate the design elements from the schematic to an empty or partially populated layout canvas.



In the **Generate Layout** form, in **Generate** tab, keep the **In Boundary** option **ON**. In the **PR Boundary** tab, keep **Utilization (%)** as **15** and **Aspect Ratio (W/H)** as **1**.

105 © Cadence Design Systems, Inc. All rights reserved.

cadence

You can generate all from the source by choosing either the Connectivity – Generate – All From Source menu command or clicking the Generate All From Source icon on the Layout EXL toolbar.

The Generate Layout form appears.

The GFS command populates the design elements from the schematic to an empty or partially populated layout canvas.

- In the Generate Layout form, in Generate tab, keep *In Boundary* option **ON**.
- In PR Boundary tab, keep *Utilization (%)* as **15**, and *Aspect Ratio (W/H)* as **1**, and do the placement.

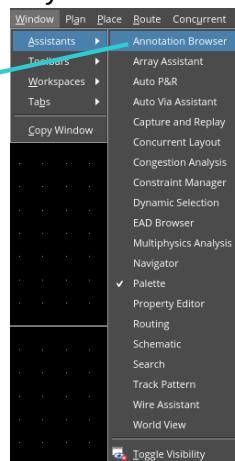
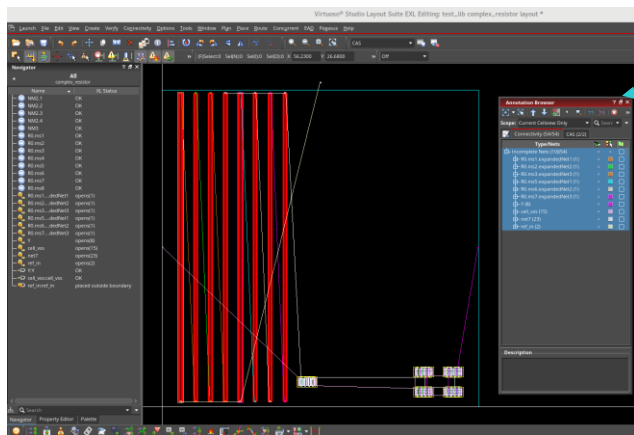
Using the Annotation Browser to Display the Flight Lines



Invoke the Annotation Browser assistant by choosing either the **Window – Assistants – Annotation Browser** pull-down menu or clicking the **Annotation Browser** icon on the Layout EXL toolbar.



Use the Annotation Browser assistant to display the flight lines of the layout.



106 © Cadence Design Systems, Inc. All rights reserved.

cadence

You can invoke the Annotation Browser assistant by choosing either the **Window – Assistants – Annotation Browser** pull-down menu or by clicking the **Annotation Browser** icon on the Layout EXL toolbar.

As expected, eight resistors of 5 Ohms each have been generated.

You can display the flight lines in the layout using the Annotation Browser assistant to see how they are connected.

There are two groups in parallel, each group including four resistors in series.

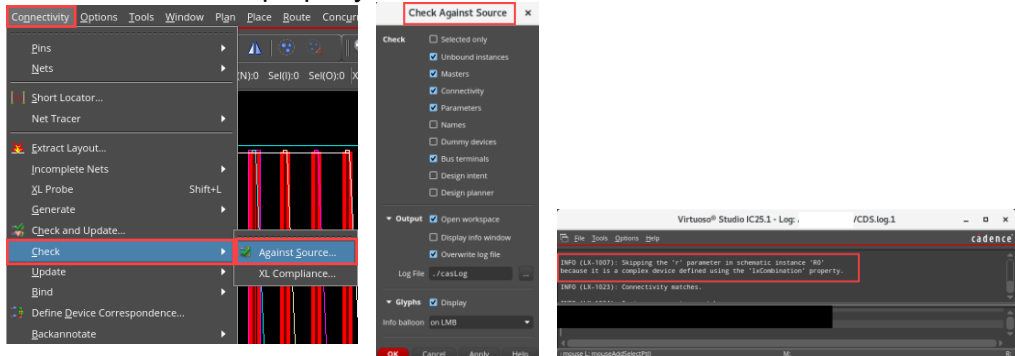
Checking Against Source



Execute Check Against Source by choosing either **Connectivity – Check – Against Source** or by clicking the **Check Against Source (CAS)** icon on the Layout EXL toolbar.



The “r” parameter in schematic instance “R0” is skipped because it is a complex device defined using the “*lxCombination*” property.



Check Against Source (CAS) generates a summary report (Info).

107 © Cadence Design Systems, Inc. All rights reserved.

cadence

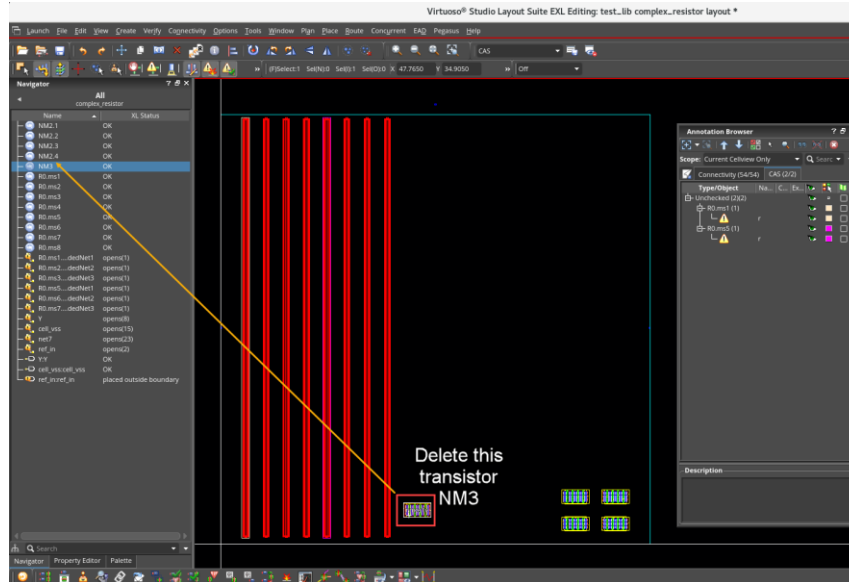
You can execute the Check Against Source command by choosing either the **Connectivity – Check – Against Source** pull-down menu or clicking the **Check Against Source** icon on the Layout EXL toolbar.

After running the command, Check Against Source generates a summary report.

You can observe that the “r” parameter in schematic instance “R0” is skipped because it is a complex device defined using the “*lxCombination*” property.

Checking Against Source Deleting an Instance in the Layout

Delete the layout instance **NM3** as shown here:



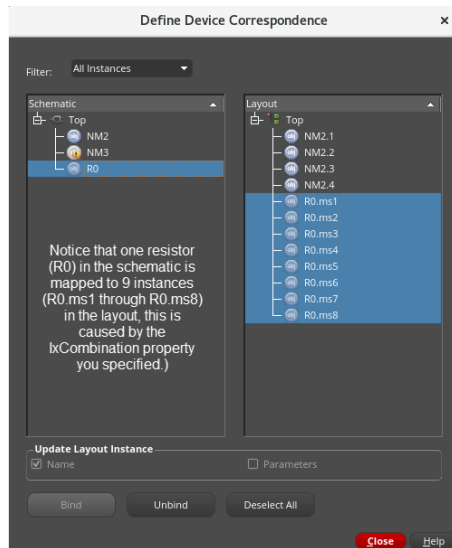
108 © Cadence Design Systems, Inc. All rights reserved.



You have deleted the layout instance *NM3*.

Checking Against Source Mapping in Define Device Correspondence

The schematic instance **NM3** is now unmapped, hence you see the flag close to its name.



109 © Cadence Design Systems, Inc. All rights reserved.



The schematic instance **NM3** is now unmapped, hence you see the flag close to its name.

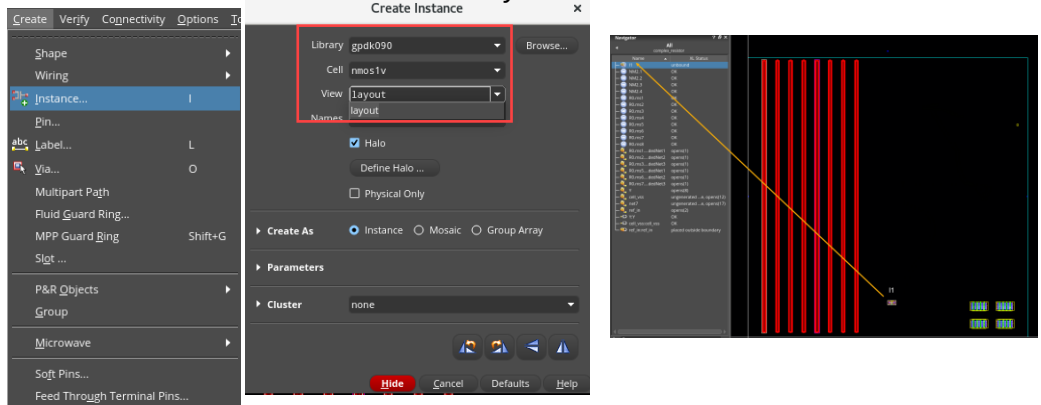
Creating an Instance



Invoke the Create Instance form by choosing either the **Create – Instance** menu command or pressing the bindkey **I** or clicking the **Create Instance** icon on the Create toolbar.



Use the Create Instance command to instantiate a new layout instance.



A new layout instance *gpdK090/nmos1v/layout* is instantiated in the design.

110 © Cadence Design Systems, Inc. All rights reserved.



You can invoke the Create Instance form by choosing either the **Create – Instance** menu command or pressing the bindkey **I** or clicking the **Create Instance** icon on the Create toolbar.

You can use the Create Instance command to instantiate a new layout instance.

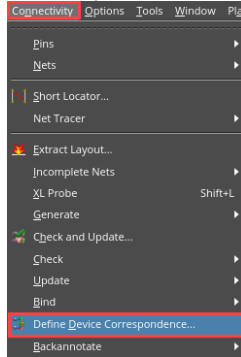
A new layout instance *gpdK090/nmos1v/layout* is instantiated in the layout design.

Defining the Device Correspondence



Run Define Device Correspondence by choosing either **Connectivity – Define Device Correspondence** menu command or clicking the **Define Device Correspondence (DDC)** icon on the Layout EXL toolbar.

- Use the Define Device Correspondence form to create or update the binding between schematic and layout instances or terminals for which correspondence is defined.
- Device correspondence also propagates connectivity of the wiring connections between the schematic and the layout.



111 © Cadence Design Systems, Inc. All rights reserved.



You can run Define Device Correspondence command by choosing either *Connectivity – Define Device Correspondence* menu command or by clicking the *Define Device Correspondence* icon on the Layout EXL Toolbar.

You can use the Define Device Correspondence form to create or update the binding between schematic and layout instances or terminals for which correspondence is defined.

Device correspondence also propagates the connectivity of the wiring connections between the schematic and the layout.

Binding Schematic/Layout Using DDC

- There is one unmapped instance (**NM3**) in the schematic and one in the layout (**I1**).
- Select both the instances **NM3** and **I1** and then click the **Bind** button.

The image shows two screenshots of the 'Define Device Correspondence' (DDC) dialog box. The left screenshot shows the initial state where 'NM3' is selected in the Schematic pane and 'I1' is selected in the Layout pane. The right screenshot shows the state after the 'Bind' button is clicked, where 'NM3' is now selected in both the Schematic and Layout panes.

Annotations:

- Select All Instances in the Filter cyclic field to also show the already mapped ones.** (Points to the 'All Instances' filter dropdown in the left screenshot.)
- Updates the layout instance name to match the corresponding schematic instance.** (Points to the 'Update Layout Instance' checkbox in the left screenshot.)
- Creates a correspondence between the devices selected in the Schematic pane and the devices selected in the Layout pane. This button is enabled only if all the selected devices are unbound.** (Points to the 'Bind' button in the left screenshot.)
- The schematic /layout binding has now been established for NM3** (Points to the 'NM3' instance in the Layout pane in the right screenshot.)

DDC Dialog Details:

- Filter:** All Instances
- Schematic Pane:**
 - Top
 - NM2
 - NM3 (Selected)
 - R0
- Layout Pane:**
 - Top
 - I1 (Unmapped Instance)
 - NM2.1
 - NM2.2
 - NM2.3
 - NM2.4
 - R0.ms1
 - R0.ms2
 - R0.ms3
 - R0.ms4
 - R0.ms5
 - R0.ms6
 - R0.ms7
 - R0.ms8
- Buttons:** Bind, Unbind, Deselect All, Close, Help
- Update Layout Instance:**
 - ☒ Name
 - ☐ Parameters

The schematic/layout binding has now been established for **NM3** as seen in DDC.

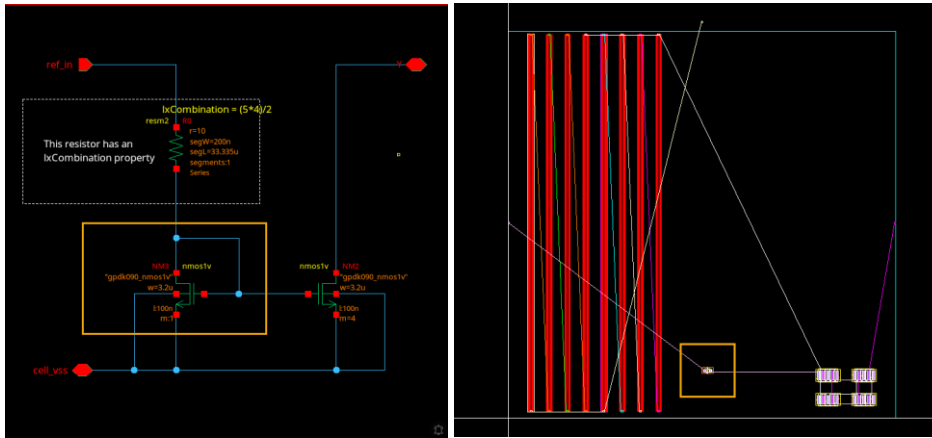
Let us see how to bind the schematic and layout instances using the *Define Device Correspondence* command.

There is one unmapped instance *NM3* in the schematic and one unmapped instance *I1* in the layout. Select both the instances *NM3* and *I1* and then click the *Bind* button.

The schematic and layout binding has now been established for *NM3* as seen in the DDC form.

An Example of Cross-Probing and XL Compliance after Complex Binding

- Schematic and Layout are with XL Compliance.
- The layout instance names match the schematic instance names.



The schematic/layout binding has now been established and you can cross-select the instances in schematic/layout.

113 © Cadence Design Systems, Inc. All rights reserved.



An Example of Cross-Probing and XL Compliance After Complex Binding is given here.

Cross-probing lets you select an object in the *layout* and the software subsequently selects and highlights that object in the *schematic*, and vice versa.

Also, an object selected in the *Navigator*, *Annotation Browser*, or *Constraint Manager* will cross-select and highlight as well.

Because this is connectivity driven, this feature is available in the *XL* and *EXL* and *MXL* suite.

Schematic and Layout views with EXL Compliance is shown here.

You can see that the layout instance names match to that of the schematic instance names.

The schematic/layout binding has now been established, and you can cross-select the instances in schematic/layout.

Module Summary

In this module, you learned to

- Analyze how the Virtuoso Layout-EXL Binder and Extractor work
- Debug connectivity problems
- Define advanced Series/Parallel mapping
- Display the connectivity information using the Annotation Browser assistant and the Navigator assistant



In this module, you learned to analyze how the Virtuoso Layout-EXL Binder and Extractor work.

You debugged the connectivity problems.

You defined the advanced Series/Parallel mapping.

You displayed the connectivity information using the Annotation Browser Assistant and the Navigator Assistant.

Labs



- Lab 3-1 Using Manual Binding
- Lab 3-2 Using Incremental Binding
- Lab 3-3 Using Update Binding
- Lab 3-4 Using Complex Binding



Now, you will do Lab Exercises in this Module 3.

There are 4 labs in this module.

- In the first lab, *Manual Binding*, you bind the layout instances with their schematic counterpart by Manual Binding using the Define Device Correspondence command.
- In the second lab, *Incremental Binding*, you bind the layout instances with their schematic counterpart using the Automatic Incremental Binder and the Update Components and Nets command.
- In the third lab, *Update Binding*, you consider a design which was completed outside of Layout EXL and make it XL compliant using the Update Binding command.
- In the fourth lab, *Complex Binding*, you explore the Series/Parallel mapping and the Complex Binding feature.

Quiz: Binding Between Schematic/Layout



When you have pins and instances correctly bound between the schematic and the layout in the design, it is called _____.

(Enter the answer in the blank field.)

Answer: *XL Compliance*

XL Compliance is having pins and instances correctly bound between the schematic and the layout in the design.



This page does not contain notes.

Quiz: Highlighting/Referencing in the Design



Selected objects will highlight/reference in the schematic, Navigator, Annotation Browser, Constraint Manager, Property Editor, and layout when you:

(Select the single correct choice.)

- A. Edit In Place
- B. Descend Edit
- C. Update Binding
- D. Cross Probe
- E. None of the above

Answer: D

Selected objects will highlight/reference in the schematic, Navigator, Annotation Browser, Constraint Manager, Property Editor, and layout when you *Cross Probe*.



This page does not contain notes.

Quiz: Create/Update Binding Between Schematic/Layout



Which form can be used to create or update the binding between the schematic and the layout instances or terminals in the design?

(Select the single correct choice.)

- A. Define Device Correspondence (DDC)
- B. Update Binding
- C. Update Components and Nets (UCN)
- D. Check Against Source (CAS)
- E. None of the above

Answer: A

You can use the Define Device Correspondence (DDC) form to create or update the binding between the schematic and the layout instances or terminals in the design.



This page does not contain notes.

Quiz: Finding the Differences Between Schematic/Layout



Which command can be used to find the differences between the schematic and the layout in the design?

(Select the single correct choice.)

- A. Define Device Correspondence (DDC)
- B. Update Binding
- C. Update Components and Nets (UCN)
- D. Check Against Source (CAS)
- E. None of the above

Answer: D

You can use the Check Against Source (CAS) command to find the differences between the schematic and the layout in the design.



This page does not contain notes.

Quiz: Cross-Probing in Layout Design



Cross-Probing is available in all the Tiers, i.e., in XL, EXL and MXL Suites in the layout design.
(Select the single correct choice from the two choices.)

- A. True
- B. False

Answer: A

True. Because *Cross-Probing* is connectivity driven, this feature is available in XL, EXL and MXL Suites in the layout design.



This page does not contain notes.



Module 4

Using Constraint Groups in Virtuoso®

cadence®

This page does not contain notes.

Module Objectives

In this module,

- Define the concept of Constraint Groups
- Examine the Constraint Groups that affect the way Layout EXL extracts the connectivity
- Illustrate how to select proper Constraint Groups



In this module, you describe the concept of Constraint Groups.

You explore how to define the Constraint Groups.

You analyze how the Constraint Groups affect the way Layout EXL extracts the connectivity.

You discuss how to select the proper Constraint Groups for your design.

What Is a Constraint Group?



A **Constraint Group (CG)** lets you specify the different sets of design constraints to be applied under different design circumstances or for different design objects. This allows the flexibility to experiment with more or less stringent process rules at different stages of the design process.

- You can group together any set of constraints into a Constraint Group, which can then be applied to any design associated with the technology library.
- The foundry constraint group represents the absolute minimum rules (i.e., *minWidth/minSpacing*, etc.) which must be adhered to in order to manufacture the design. These are commonly referred to as the base process rules and are stored only in the technology database.



A Constraint Group (CG) lets you specify the different sets of design constraints to be applied under different design circumstances or for different design objects. This allows the flexibility to experiment with more or less stringent process rules at different stages of the design process.

You can group together any set of constraints into a Constraint Group, which can then be applied to any design associated with the technology library.

The *foundry* constraint group represents the absolute minimum rules (i.e., *minWidth/minSpacing*, etc.) which must be adhered to in order to manufacture the design. These are commonly referred to as the base process rules and are stored only in the technology database.

For more information, refer to the *Constraint Groups and Constraints* in the *Virtuoso Technology Data Constraints Reference, Product Version IC23.1*.

Constraint Group: Describing Your Technology

Design Rules in a techfile are shown here in the layout view.

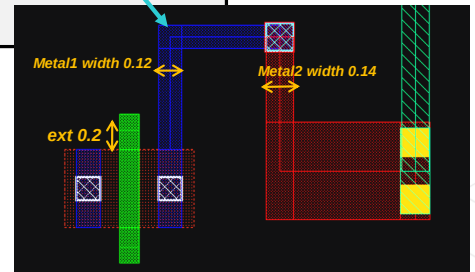
```
; min width rules
spacings(
  ( minWidth    "Poly"    0.100 )
  ( minWidth    "Metal1"  0.120 )
  ( minWidth    "Metal2"  0.140 )
) ; spacings

; example of table based rule
spacingTables(
  ( minNumCut   "Cont"
    (( "width" nil nil ) 1 )
    ( 0.12 1
      0.4 2 ; metal > 400n => 2 cuts
      1.0 4 ; metal > 1u  => 4 cuts
    )
  )
) ; spacingTables

orderedSpacings(
  ( minExtensionDistance "Metal1" "Cont" 0.0 )
  ( minExtensionDistance "Oxide"  "Poly"  0.2 )
) ; orderedSpacings
```

```
; no shape-drawn transistor is allowed
techDerivedLayers(
  ( noOverlapLayer2 10001 ( Poly 'and Resdum ) )
) ; techDerivedLayers

; description of the allowed routing layers/vias
interconnect(
  ( validLayers (Poly Metal1 Metal2 Metal3 Metal4))
  ( validVias   (M1_P0 M2_M1 M3_M2 M4_M3))
  ( errorLayer  noOverlapLayer1 )
  ( errorLayer  noOverlapLayer2 )
) ; interconnect
```



124 © Cadence Design Systems, Inc. All rights reserved.

cadence

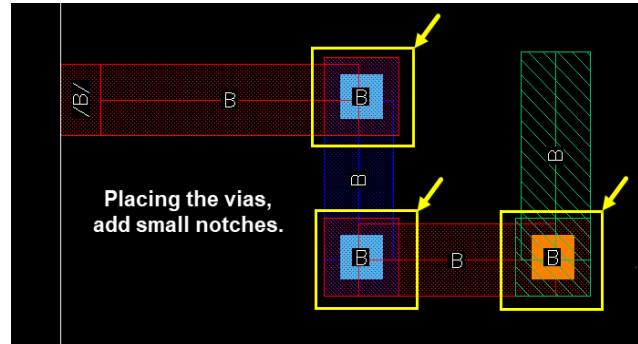
Design Rules in a techfile are shown here in the layout view.

Routing: Constraint Group – Basic Set of Rules



Foundry CG with a basic set of rules

```
( "foundry" nil
  spacings(
    ( minWidth "Poly" 0.100 )
    ( minWidth "Metal1" 0.120 )
    ( minWidth "Metal2" 0.140 )
  ) ; spacings
) ; foundry
```



A Constraint Group provides a description of your technology, defining the design rules, describing the mask-stack, providing the electrical properties of the layers and defining how they must be treated in Virtuoso.

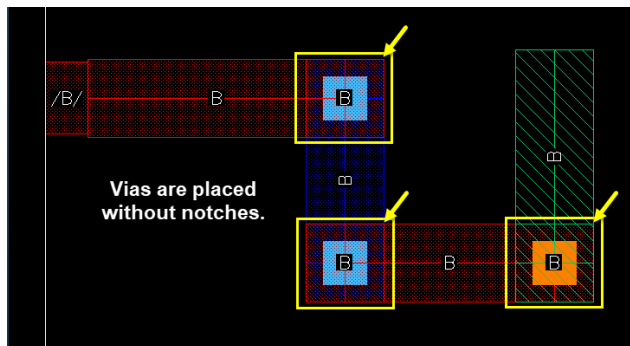
Every techfile must include a Constraint Group (CG) named *foundry*, which defines the basic set of rules which *cannot be violated*.

Routing: Constraint Group – Relaxing Minimum Rules



CleanRouting CG with relaxed rules

```
( "CleanRouting" nil
  spacings(
    ( minWidth "Poly" 0.120 )
    ( minWidth "Metal1" 0.260 )
    ( minWidth "Metal2" 0.260 )
    ( minWidth "Metal3" 0.260 )
  ) ; spacings
) ; CleanRouting
```



Other Constraint Groups, for example *CleanRouting*, may “relax” some of the rules and/or reduce the choice of allowed layers/vias. In the example presented in this foil, the *minWidth* of *Poly*, *Metal1*, *Metal2*, and *Metal3* layers have been increased to avoid notches when placing the vias.

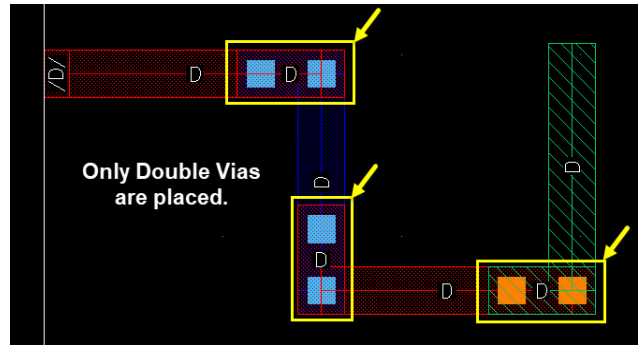
The Constraint Group *CleanRouting* does not reflect the technology design rules and leads to unnecessary *Wide Poly* and *Metal2* lines but leads to an increased manufacturability.

Routing: Constraint Group – Placing DoubleVias



With **DoubleVias** as **Constraint Group**, only double vias are placed during the interactive routing.

```
( "DoubleVias"      nil
  spacingTables(
    ( minNumCut "Cont" (( "width" nil nil)) (0.0 2))
    ( minNumCut "Via1" (( "width" nil nil)) (0.0 2))
    ( minNumCut "Via2" (( "width" nil nil)) (0.0 2))
    ( minNumCut "Via3" (( "width" nil nil)) (0.0 2))
  ) ;spacingTables
) ; DoubleVias
```



With DoubleVias as the Constraint Group, only double vias are placed during the interactive routing.

Module Summary

In this module, you learned to

- Define the concept of Constraint Groups
- Analyze how the Constraint Groups affect the way Layout EXL extracts the connectivity
- Illustrate how to select proper Constraint Groups



This page does not contain notes.

Lab



Lab 4-1 Using Constraint Groups

129 © Cadence Design Systems, Inc. All rights reserved.



Now, you will do the lab exercise in this Module 4 on Constraint Groups.

You explore how to define the Constraint Groups and how to select the proper Constraint Group for your design.

Quiz: Constraint Group in the Technology File



Every techfile must include a Constraint Group (CG); this CG defines the basic set of rules that cannot be violated.

(Select the single correct choice.)

- A. virtuosoDefaultSetup
- B. foundry
- C. LEFDefaultRouteSpec
- D. virtuosoDefaultExtractorSetup
- E. None of the above

Answer: B

Every techfile must include a Constraint Group (CG) named *foundry*; this CG defines the basic set of rules that cannot be violated.



This page does not contain notes.

Quiz: Constraints in Layout Design



Constraints can be created, edited, or deleted in all the Tiers, i.e, in XL, EXL and MXL Suites in the layout design.

(Select the single correct choice from the two choices.)

- A. True
- B. False

Answer: B

False. Constraints can *ONLY* be created, edited, or deleted in the Virtuoso® EXL/MXL Suites in the layout design.



This page does not contain notes.

Quiz: Grouping Constraints in the Design



You can group together any set of constraints, which can then be applied to any design associated with the technology library which is called _____.

(Enter the answer in the blank field.)

Answer: *Constraint Group*

You can group together any set of constraints into a *Constraint Group*, which can then be applied to any design associated with the technology library.



This page does not contain notes.

Quiz: Utility to Load/Attach/Dump the Technology File of the Design



Which utility can be used to Load/Attach/Dump the technology file of the design?
(Select the single correct choice.)

- A. Technology File Manager
- B. Technology System Manager
- C. Technology Utility Manager
- D. Technology Manager
- E. None of the above

Answer: A

Technology File Manager can be used to Load/Attach/Dump the technology file of the design.



This page does not contain notes.

Quiz: Default Wire Constraint Group for Creating Wire



Which form can be used to select the *Default Wire Constraint Group* for creating wire?
(Select the single correct choice.)

- A. Display Options
- B. Create Wire
- C. Create Bus
- D. Layout Editor Options
- E. None of the above

Answer: D

The *Layout Editor Options* form can be used to select the *Default Wire Constraint Group* for creating wire.



This page does not contain notes.



Module 5

Course Conclusions

cādence[®]

This page does not contain notes.

Summary

In this course, you learned to

- Use Virtuoso® Layout Suite (LS) EXL to create a new layout cell from scratch
- Debug and address connectivity errors in your layout design
- Analyze how the Layout-EXL binder and extractor work
- Describe how the Constraint Groups affect working with VLS-EXL



This page does not contain notes.

References

Refer to the given links for more details on the topics in this course.

- [Virtuoso Layout Suite XL: Connectivity Driven Editing User Guide IC25.1](#)
 - This document describes how to use the Virtuoso Layout Suite XL Layout Editor (Layout XL) in a connectivity-based environment.
- [Virtuoso Automated Device Placement and Routing Flow Guide IC25.1](#)
 - The Virtuoso Layout Suite EXL environment enables you to accomplish advanced tasks for layout, including analog automatic placement, the creation of complex placement structures (module generators), and cell planning.
- [Virtuoso Custom Digital Placer User Guide IC25.1](#)
 - This document describes how to use the Virtuoso Layout Suite EXL custom digital placer.
- [Virtuoso Space-based Router User Guide IC25.1](#)
 - The Virtuoso Space-based Router GUIs aids designers in using different routing features.
- [Virtuoso Layout Viewer User Guide IC25.1](#)
 - Layout Viewer provides various assistants to enable you to work on your designs effectively and efficiently.



[Virtuoso Layout Suite XL: Connectivity Driven Editing User Guide IC25.1](#)

[Virtuoso Automated Device Placement and Routing Flow Guide IC25.1](#)

[Virtuoso Custom Digital Placer User Guide IC25.1](#)

[Virtuoso Space-based Router User Guide IC25.1](#)

[Virtuoso Layout Viewer User Guide IC25.1](#)



Module 6

Next Steps

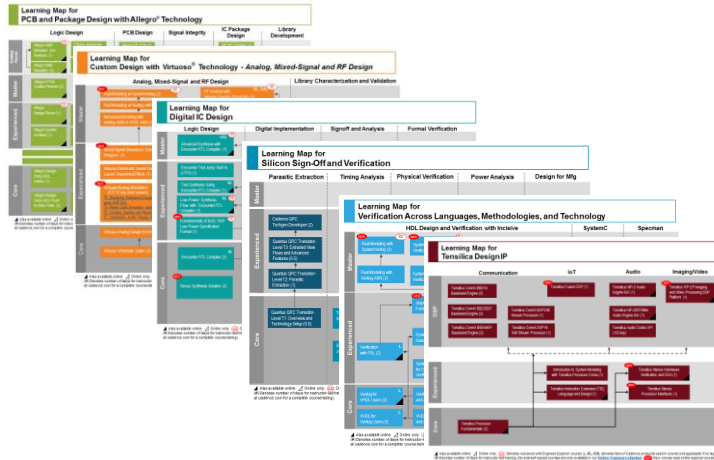
cādence[®]

This page does not contain notes.

Learning Maps

Cadence® Training Services learning maps provide a comprehensive visual overview of the learning opportunities for Cadence customers.

Click [here](#) to see all our courses in each technology area and the recommended order in which to take them.



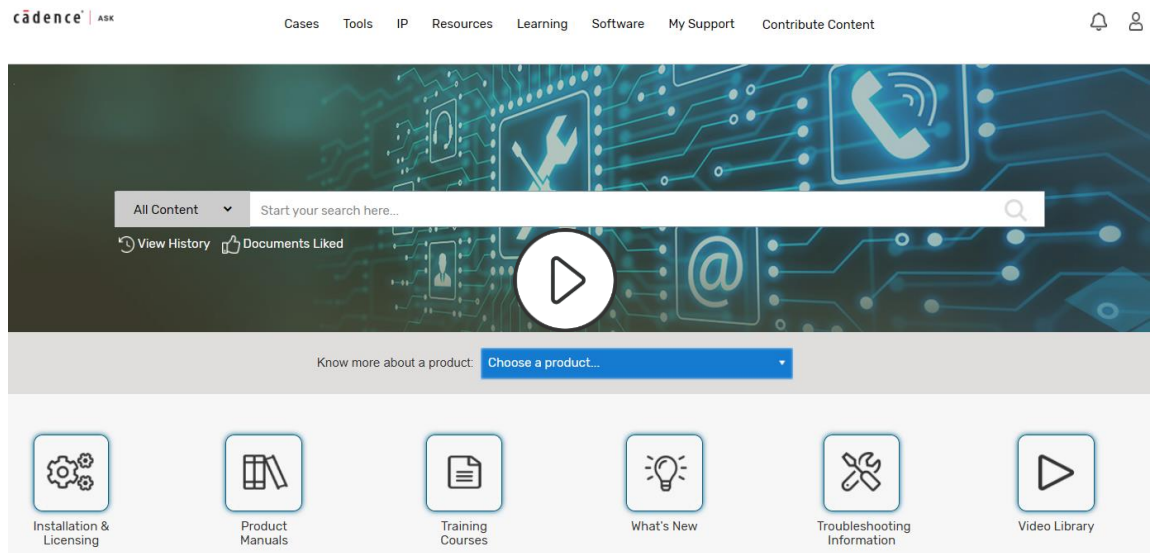
139 © Cadence Design Systems, Inc. All rights reserved.



Go here to view the learning maps:

http://www.cadence.com/Training/Pages/learning_maps.aspx

Cadence ASK – Learning and Support Portal



[Cadence Ask](#) now includes over 2000 product/language/methodology videos (“Training Bytes”)!

140 © Cadence Design Systems, Inc. All rights reserved.



Click [here](#) to view a demo of Welcome to Cadence ASK – Learning and Support Portal.

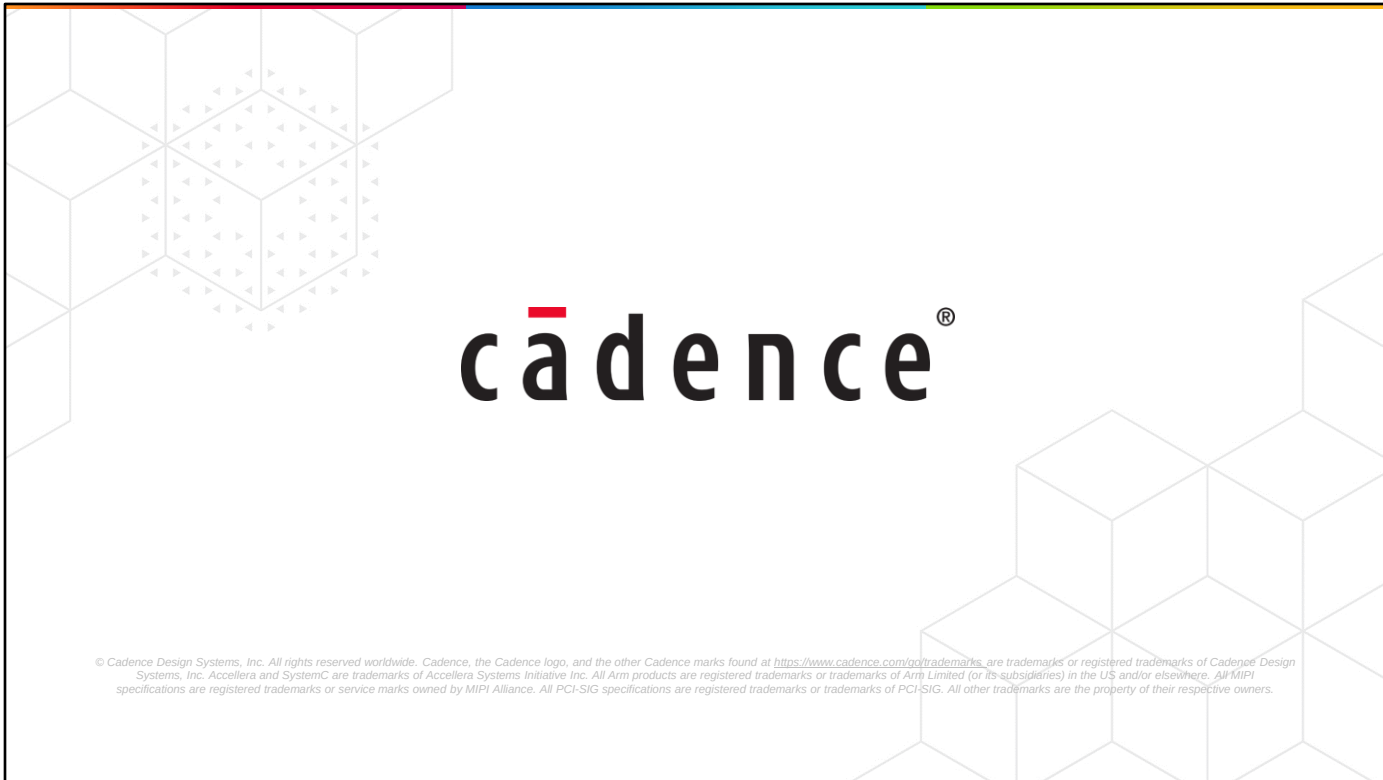
Wrap Up

- Complete Post Assessment, if provided
- Complete the Course Evaluation
- Get a Certificate of Course Completion

Thank you!



This page does not contain notes.



© Cadence Design Systems, Inc. All rights reserved worldwide. Cadence, the Cadence logo, and the other Cadence marks found at <https://www.cadence.com/oc/trademarks> are trademarks or registered trademarks of Cadence Design Systems, Inc. Accellera and SystemC are trademarks of Accellera Systems Initiative Inc. All Arm products are registered trademarks or trademarks of Arm Limited (or its subsidiaries) in the US and/or elsewhere. All MIPI specifications are registered trademarks or service marks owned by MIPI Alliance. All PCI-SIG specifications are registered trademarks or trademarks of PCI-SIG. All other trademarks are the property of their respective owners.

This page does not contain notes.